From the Normal Equation to PCA: Eigen-directions

ES 335 / ES 678 Machine Learning, Sem I 2026 Source: `sources/course-2026/lectures/L05_normal_equation_eigendirections_pca.pdf` (35 slides) · Previous: Loss and projection

How to use this file. Part A builds the *new* maths Lecture 5 needs, on top of Lectures 1–4 (gradients of matrix expressions, Gram matrix $X^\top X$, PSD, rank, projection). Most of it is **eigenvalues and eigenvectors**, the **spectral theorem**, **covariance** and **Lagrange multipliers**. Part B goes through all 35 slides. The "work it out" slides are solved in this PDF; here every step is expanded and checked with numbers, and extra worked examples are added where the slide only gives symbols. Part C is practice.

What this lecture is really about: one matrix question, asked twice.

$$\underbrace{X^\top X\hat{\theta} = X^\top y}_{\text{normal equation}} \xrightarrow{X^\top X = Q\Lambda Q^\top} \underbrace{\lambda_j a_j = b_j}_{\text{one scalar equation per direction}} \qquad \Big| \qquad \underbrace{\max_{q^\top q = 1} q^\top C q}_{\text{PCA}} \xrightarrow{\text{Lagrange}} \underbrace{Cq = \lambda q}_{\text{same eigen-algebra}}$$

In **regression**, eigenvalues of $X^\top X$ say how strongly the data pin down each coefficient direction (big λ = steep, stable; small λ = flat, sensitive). In **PCA**, eigenvalues of the covariance C say how much the data *vary* along each feature direction (big λ = keep; small λ = cheap to drop).

Step-up path

You need. The normal equations, and what a matrix does to a vector.

1. Hold two pictures at once: coefficient space and observation space.
2. Find the directions a symmetric matrix only stretches, its eigenvectors.
3. Read the loss bowl through those directions: eigenvalues are curvatures.
4. Connect small eigenvalues to unstable coefficients.
5. Maximise variance along a direction, which gives PCA as an eigenvalue problem.

You can stop when you can explain why near-duplicate columns make coefficients unstable, using eigenvalues.

Contents

Part A — Maths from the ground up

1. Two spaces: coefficient space vs observation space
2. The gradient, one coordinate at a time
3. The Gram matrix $X^\top X$
4. Eigenvalues and eigenvectors
5. Orthonormal vectors, orthogonal matrices, change of basis
6. The spectral theorem and quadratic forms
7. Condition number and sensitivity
8. Centring, variance and the covariance matrix
9. Projecting onto a single unit direction
0. Lagrange multipliers
1. Frobenius norm, trace and low-rank reconstruction

Part B — Slides · All 35 slides

Part C — Practice · Jump

Part A Maths From the Ground Up

A1 Two spaces: coefficient space vs observation space

Lecture 5's main trap is mixing up **which space a vector lives in**. Keep two spaces apart:

Space	Dimension	What lives there	Examples
coefficient / feature space	$\mathbb{R}^d$ (one axis per predictor)	θ, ∇J, eigenvectors q_j of $X^\top X$, PCA directions q	$\theta = (2.2, 2.2)$
observation space	$\mathbb{R}^n$ (one axis per data row)	y, $\hat{y} = X\theta$, r, PCA scores $z = X_c q$	$y = (2, 5, 6, 9)$

$X \in \mathbb{R}^{n \times d}$ is the bridge: X maps $\mathbb{R}^d \to \mathbb{R}^n$, and $X^\top$ maps back $\mathbb{R}^n \to \mathbb{R}^d$. So:

- $X^\top r$ is a d-vector (one number per coefficient) → it can be a gradient.
- $X^\top X$ is $d \times d$ → it acts on coefficient moves.
- Xq is an n-vector → it's the *prediction pattern* produced by moving coefficients along q.

Shape check is a correctness check (L2). If a formula adds an n-vector to a d-vector, it's wrong.

A2 The gradient, one coordinate at a time

L4 A9 gave the matrix rule $\nabla_\theta \frac{1}{2}\|y - X\theta\|^2 = X^\top(X\theta - y)$. Lecture 5 re-derives it *without matrix calculus*, using only the scalar chain rule.

Write row i out: $r_i(\theta) = y_i - \sum_{k=1}^{d} x_{ik}\theta_k$. Only the $k = j$ term contains θ_j, so

$$\frac{\partial r_i}{\partial \theta_j} = -x_{ij}.$$

Chain rule on $J = \frac{1}{2}\sum_i r_i^2$:

$$\frac{\partial J}{\partial \theta_j} = \sum_i r_i \frac{\partial r_i}{\partial \theta_j} = -\sum_i x_{ij} r_i.$$

$\sum_i x_{ij} r_i$ is "column j of X, dotted with r" $= (X^\top r)_j$. Stack $j = 1..d$: $\nabla J = -X^\top r = X^\top(X\theta - y)$.

Why the transpose? The sum runs over rows i (the n direction), and the answer must be indexed by j (the d direction). Summing over the *first* index of X is exactly what $X^\top$ does.

A3 The Gram matrix $X^\top X$

Write X by columns, $X = [x_1 \ \cdots \ x_d]$, each $x_j \in \mathbb{R}^n$ = predictor j measured on all n rows. Then entry (j, k) of $X^\top X$ is a dot product of two columns:

$$(X^\top X)_{jk} = x_j^\top x_k.$$

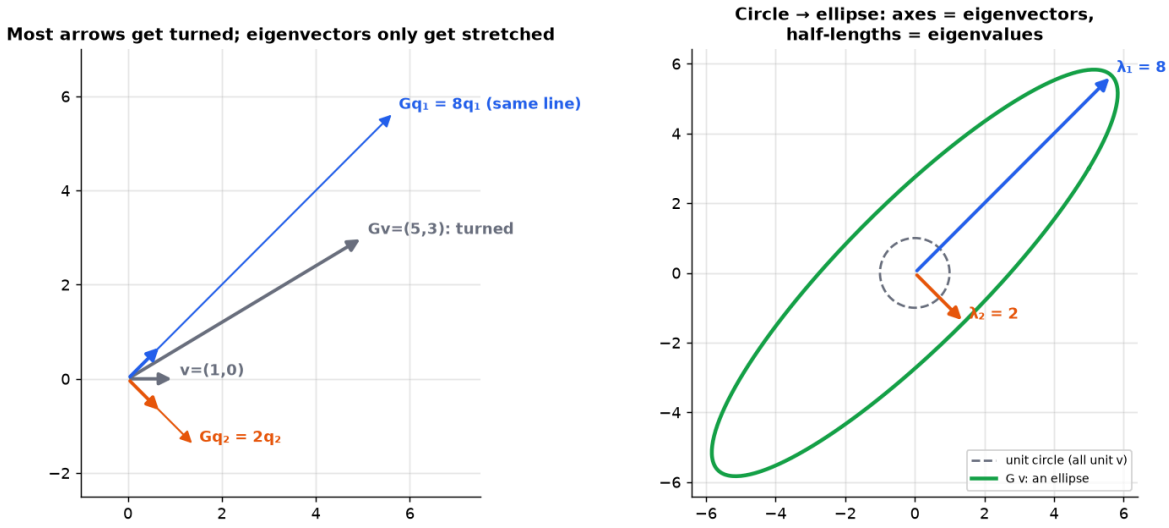
Entry	Meaning
diagonal $x_j^\top x_j = \|x_j\|^2$	the squared size of predictor j
off-diagonal $x_j^\top x_k$	how much predictors j and k **overlap** (point the same way across rows)

A matrix of pairwise dot products is called a **Gram matrix**. Properties (from L4 A10):

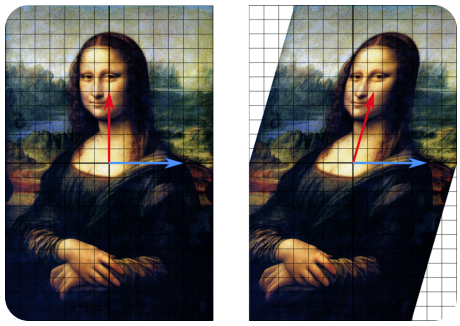
- **symmetric**: $x_j^\top x_k = x_k^\top x_j$;
- **PSD**: $v^\top X^\top X v = \|Xv\|^2 \geq 0$;
- **PD** iff full column rank.

If columns are centred, $X^\top X/(n-1)$ is the **covariance matrix** (A8): diagonal = variances, off-diagonal = covariances. Strongly correlated predictors ⇒ large off-diagonal entries ⇒ "coupling".

A4 Eigenvalues and eigenvectors



G = [[5,3],[3,5]]: (1,0) gets turned, q_1 and q_2 only stretch (×8, ×2). The unit circle becomes an ellipse along them.



A shear moves the red arrow off its line; the blue arrow (an eigenvector) stays on its line. (Wikimedia Commons: Mona_Lisa_eigenvector_grid.png, CC0)

Most matrices **rotate and stretch** a vector. A few special directions only get **stretched**:

$$Gq = \lambda q, \qquad q \neq 0.$$

q is an **eigenvector** (a direction the matrix keeps on the same line), λ its **eigenvalue** (the stretch factor; negative means flipped, zero means squashed to 0).

How to find them. $Gq = \lambda q \iff (G - \lambda I)q = 0$. A nonzero q solving this exists only if $G - \lambda I$ is singular (det = 0, it flattens the plane; see the matrix toolkit in L04 §A7):

$$\det(G - \lambda I) = 0 \qquad (\text{characteristic equation}).$$

Solve for λ, then for each λ solve $(G - \lambda I)q = 0$ for q.

2×2 shortcuts:

Fact	Formula
sum of eigenvalues	$\lambda_1 + \lambda_2 = \text{trace}(G) = a + d$
product of eigenvalues	$\lambda_1 \lambda_2 = \det G = ad - bc$
so	$\lambda = \frac{T}{2} \pm \sqrt{\frac{T^2}{4} - D}$ with T = trace, D = det
symmetric $\begin{bmatrix} a & b \\ b & a \end{bmatrix}$	$\lambda = a \pm b$, eigenvectors $\frac{1}{\sqrt{2}}(1,1)$ and $\frac{1}{\sqrt{2}}(1,-1)$

Example. $G = \begin{bmatrix} 5 & 3 \\ 3 & 5 \end{bmatrix}$: $\lambda = 5 \pm 3 = 8, 2$. Check $q = (1,1)$: $Gq = (8,8) = 8q$ ✓. $q = (1,-1)$: $Gq = (2,-2) = 2q$ ✓.

Scaling. If q is an eigenvector, so is cq for any $c \neq 0$. We usually pick the **unit** one: $\|q\| = 1$. (Still a sign choice: q and $-q$ are both fine.)

A5 Orthonormal vectors, orthogonal matrices, change of basis

Vectors $q_1, \ldots, q_d$ are **orthonormal** if $q_j^\top q_k = 1$ when $j = k$ and 0 otherwise (unit length, mutually perpendicular). Stack them as columns of $Q = [q_1 \ \cdots \ q_d]$. Then

$$Q^\top Q = I \quad \text{and (square } Q) \ QQ^\top = I, \qquad \text{so } Q^{-1} = Q^\top.$$

Such a Q is an **orthogonal matrix**: a rotation (possibly with a reflection). It preserves lengths: $\|Qa\| = \|a\|$.

Change of basis. Any $\theta \in \mathbb{R}^d$ can be written in the new axes $q_1..q_d$:

$$a = Q^\top \theta \quad (a_j = q_j^\top \theta, \text{ the shadow of } \theta \text{ on } q_j), \qquad \theta = Qa = a_1 q_1 + \cdots + a_d q_d.$$

The arrow θ doesn't move. Only the ruler you measure it with changes.

Example. $q_1 = \frac{1}{\sqrt{2}}(1,1)$, $q_2 = \frac{1}{\sqrt{2}}(1,-1)$, $\theta = (3,1)$. Then $a_1 = \frac{4}{\sqrt{2}} = 2\sqrt{2}$, $a_2 = \frac{2}{\sqrt{2}} = \sqrt{2}$. Rebuild: $2\sqrt{2} \cdot \frac{1}{\sqrt{2}}(1,1) + \sqrt{2} \cdot \frac{1}{\sqrt{2}}(1,-1) = (2,2) + (1,-1) = (3,1)$ ✓.

A6 The spectral theorem and quadratic forms

Spectral theorem. Every **real symmetric** $G \in \mathbb{R}^{d \times d}$ has

1. all eigenvalues **real**;
2. a full set of d **orthonormal** eigenvectors (eigenvectors for different eigenvalues are automatically perpendicular);
3. the decomposition

$$G = Q\Lambda Q^\top = \sum_{j=1}^{d} \lambda_j q_j q_j^\top, \qquad \Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_d).$$

Read $Gv = Q\Lambda Q^\top v$ right to left: $Q^\top$ **measures** v in eigen-coordinates → Λ **scales** each coordinate independently → Q **returns** to the original axes. In the eigenbasis, G is just a diagonal stretch.

If G is also PSD (like $X^\top X$ or C), all $\lambda_j \geq 0$. If PD, all $\lambda_j > 0$. Number of nonzero eigenvalues $= \mathrm{rank}(G)$.

Quadratic forms in the eigenbasis. With $v = Qz$:

$$v^\top G v = z^\top Q^\top Q \Lambda Q^\top Q z = z^\top \Lambda z = \sum_j \lambda_j z_j^2.$$

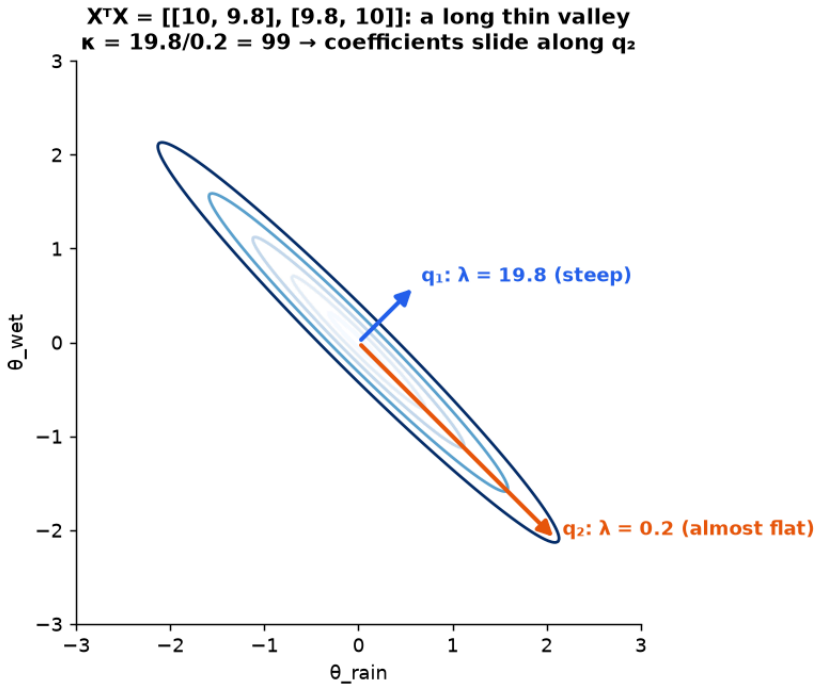
A tilted, coupled bowl becomes a sum of independent 1-D parabolas.

Rayleigh bound. For a unit vector v ($\sum z_j^2 = 1$):

$$\lambda_{\min} \leq v^\top G v = \sum_j \lambda_j z_j^2 \leq \lambda_{\max},$$

because a weighted average of the λ_j's (weights z_j^2 summing to 1) lies between the smallest and largest. **Equality at the top** when $v = q_{\max}$. This is the whole of PCA (slide 32).

A7 Condition number and sensitivity



Slide 14's matrix: steep across the valley (λ = 19.8), nearly flat along it (λ = 0.2), so the data can't pin θ down along q_2.

For a PD matrix, the **condition number** is

$$\kappa = \frac{\lambda_{\max}}{\lambda_{\min}} \quad (\geq 1).$$

$\kappa \approx 1$: round bowl, every direction equally well determined. $\kappa \gg 1$: long thin valley.

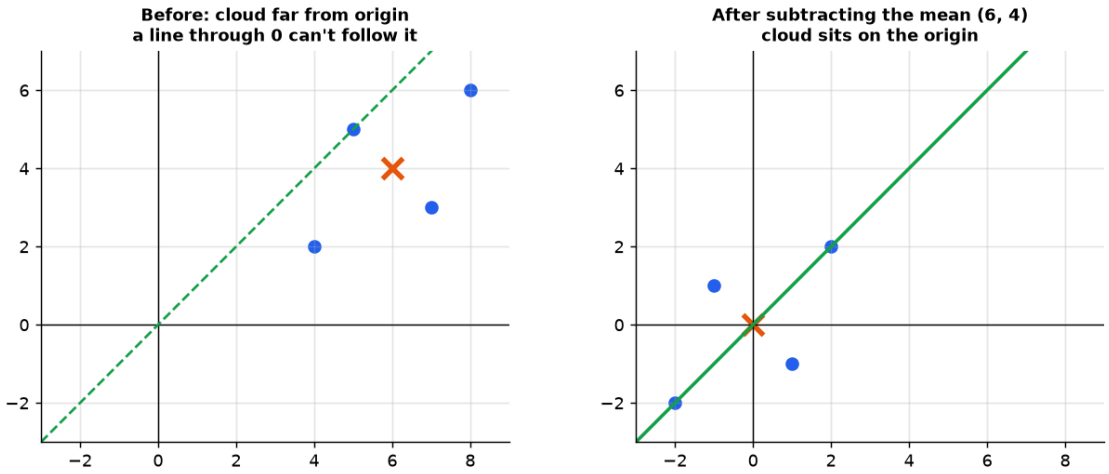
Two consequences in regression (slides 22–23):

Small λ_j means	Why
coefficient **sensitive** along q_j	$a_j = b_j/\lambda_j$: a small change in b_j gets divided by a small number
loss **flat** along q_j	curvature along q_j is λ_j
gradient descent **slow** along q_j	the gradient component is $\lambda_j z_j$: tiny push
predictions **barely change**	$\|Xq_j\|^2 = \lambda_j$: moving along q_j hardly moves $\hat{y}$

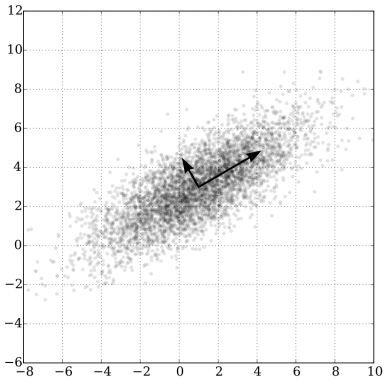
Examples. $\begin{bmatrix} 10 & 9.8 \\ 9.8 & 10 \end{bmatrix}$: $\lambda = 19.8,\ 0.2$, $\kappa = 99$. L4A's $\begin{bmatrix} 4 & 6 \\ 6 & 14 \end{bmatrix}$: $\lambda = 9 \pm \sqrt{61} \approx 16.81,\ 1.19$, $\kappa \approx 14.1$.

$\lambda_{\min} = 0$ is the extreme case: rank deficient, $\kappa = \infty$, a flat valley (L4 slide 31).

A8 Centring, variance and the covariance matrix



The toy storms before and after subtracting the mean; only after centring can a line through 0 follow the cloud.



PCA on a 2-D cloud: arrows are covariance eigenvectors, scaled by √eigenvalue. (Wikimedia Commons: GaussianScatterPCA.svg, CC BY 4.0)

Centring. Subtract each feature's mean: $x_{i,c} = x_i - \bar{x}$. Afterwards every column of X_c averages to 0: $\mathbf{1}^\top X_c = 0^\top$.

Sample variance of numbers $z_1..z_n$: $\mathrm{Var}(z) = \frac{1}{n-1}\sum_i (z_i - \bar{z})^2$. If $\bar{z} = 0$ this is $\frac{z^\top z}{n-1}$.

($n-1$, not n: the **unbiased** version, since one degree of freedom was used up estimating the mean. The argmax in PCA doesn't care which one you use: it's a positive constant.)

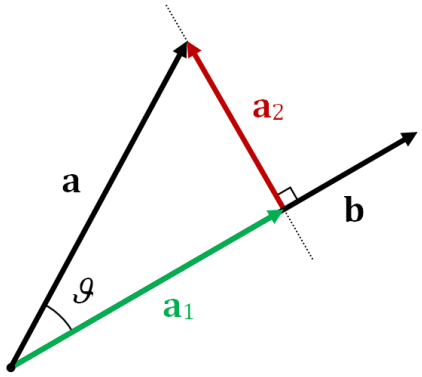
Sample covariance between features j and k: $\frac{1}{n-1}\sum_i x_{ij,c}\, x_{ik,c}$. All of them at once:

$$C = \frac{1}{n-1} X_c^\top X_c \in \mathbb{R}^{d\times d}.$$

C_{jj} = variance of feature j; C_{jk} = covariance of j and k. C is a scaled Gram matrix, so it's **symmetric PSD** (A3).

Why centre? Without it, $X^\top X/(n-1)$ measures spread around the **origin**, not around the mean. A cloud far from 0 would have a huge "variance" in the direction of its mean, even if the points barely differ.

A9 Projecting onto a single unit direction



Projecting a onto a direction b: the shadow a_1 and the perpendicular remainder a_2. (Wikimedia Commons: Projection_and_rejection.png, CC BY-SA 3.0)

Given a unit vector q ($q^\top q = 1$) and a point x, the nearest point on the line $\mathrm{span}(q)$ is zq with

$$z = q^\top x \qquad (\text{signed length of the shadow}).$$

Derivation (same perpendicular argument as L4): the leftover $x - zq$ must be perpendicular to q:

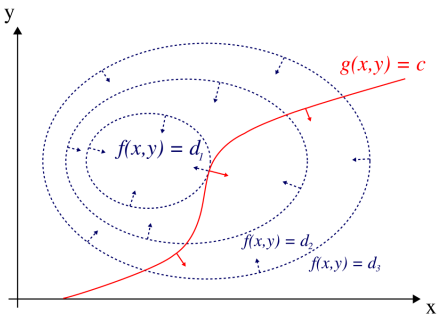
$$q^\top(x - zq) = 0 \;\Rightarrow\; q^\top x - z\underbrace{q^\top q}_{1} = 0 \;\Rightarrow\; z = q^\top x.$$

(If q weren't unit length, $z = q^\top x / q^\top q$. Fixing $q^\top q = 1$ keeps it clean.)

The projection matrix onto that line is $qq^\top$ (the 1-column case of L4's $P_X = X(X^\top X)^{-1}X^\top$ with $X = q$).

Distinguish: $q \in \mathbb{R}^d$ is a **direction** (one per dataset). $z_i \in \mathbb{R}$ is a **score** (one per observation). Stacked: $z = X_c q \in \mathbb{R}^n$.

A10 Lagrange multipliers



At the constrained optimum the contour of f is tangent to the constraint g = c, so ∇f ∥ ∇g. (Wikimedia Commons: LagrangeMultipliers2D.svg, Public domain)

Problem: maximise $f(x)$ subject to $g(x) = 0$ (you're only allowed to walk on the curve/surface $g = 0$).

Idea. At a constrained optimum you can't improve f by moving *along* the constraint. So ∇f has no component along the constraint: it points straight out of the curve, i.e. along the constraint's normal ∇g:

$$\nabla f = \lambda \nabla g \qquad \text{for some scalar } \lambda \text{ (the multiplier)}.$$

Recipe.

1. Form the **Lagrangian** $\mathcal{L}(x, \lambda) = f(x) - \lambda g(x)$.
2. Set **all** partial derivatives to zero: $\nabla_x \mathcal{L} = 0$ (gives $\nabla f = \lambda \nabla g$) and $\partial \mathcal{L} / \partial \lambda = -g = 0$ (gives back the constraint).

3. Solve. Split into cases when you divide by something that might be 0.
4. Evaluate f at **every** candidate; the largest is the max (on a closed, bounded constraint like a circle, a max is guaranteed to exist).

Mini-example. Max $x + y$ on $x^2 + y^2 = 1$. $\nabla f = (1, 1)$, $\nabla g = (2x, 2y)$. $1 = 2\lambda x$, $1 = 2\lambda y \Rightarrow x = y$. Constraint: $x = y = \pm\frac{1}{\sqrt{2}}$. Max $f = \sqrt{2}$ at $(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$.

A11 Frobenius norm, trace and low-rank reconstruction

Frobenius norm of a matrix = treat it as one long vector:

$$\|A\|_F^2 = \sum_{i,j} A_{ij}^2 = \operatorname{trace}(A^\top A) = \sum_{\text{columns}} \|a_j\|^2.$$

For data matrices, it's the **total squared error over all entries**.

Encode / decode with orthonormal columns $Q_k = [q_1 \cdots q_k] \in \mathbb{R}^{d\times k}$ ($Q_k^\top Q_k = I_k$, but $Q_k Q_k^\top \neq I_d$ unless $k = d$):

Step	Formula	Shape
encode (scores)	$Z_k = X_c Q_k$	$n \times k$
decode (reconstruction)	$\hat{X}_k = Z_k Q_k^\top = X_c\, Q_k Q_k^\top$	$n \times d$

$Q_k Q_k^\top$ is the projection onto $\operatorname{span}(q_1..q_k)$. If $k = d$ it's I, and reconstruction is perfect.

The error, using the full eigenbasis $I = \sum_{j=1}^{d} q_j q_j^\top$:

$$X_c - \hat{X}_k = X_c \sum_{j>k} q_j q_j^\top, \qquad \|X_c - \hat{X}_k\|_F^2 = \sum_{j>k} \|X_c q_j\|^2 = (n-1)\sum_{j>k} \lambda_j.$$

(The cross terms vanish because the q_j are orthonormal. And $\|X_c q_j\|^2 = (n-1) q_j^\top C q_j = (n-1)\lambda_j$.)

So **the reconstruction error is exactly the sum of the discarded eigenvalues** (times $n - 1$).

Part B Slide by Slide (all 35 slides)

Running examples used below:

- **L4A regression data** (for checks): $X = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$, $y = (2, 5, 6, 9)$, $X^\top X = \begin{bmatrix} 4 & 6 \\ 6 & 14 \end{bmatrix}$, $X^\top y = (22, 44)$, $\hat{\theta} = (2.2, 2.2)$.
- **Coupled predictors** (slide 14): $X^\top X = \begin{bmatrix} 10 & 9.8 \\ 9.8 & 10 \end{bmatrix}$.
- **Toy storms for PCA** (added here; slides 25–35 are symbolic): 4 storms, already centred, (rainfall, wetness):

storm i	1	2	3	4
centred rainfall	2	1	−1	−2
centred wetness	2	−1	1	−2

Slide 1 Title

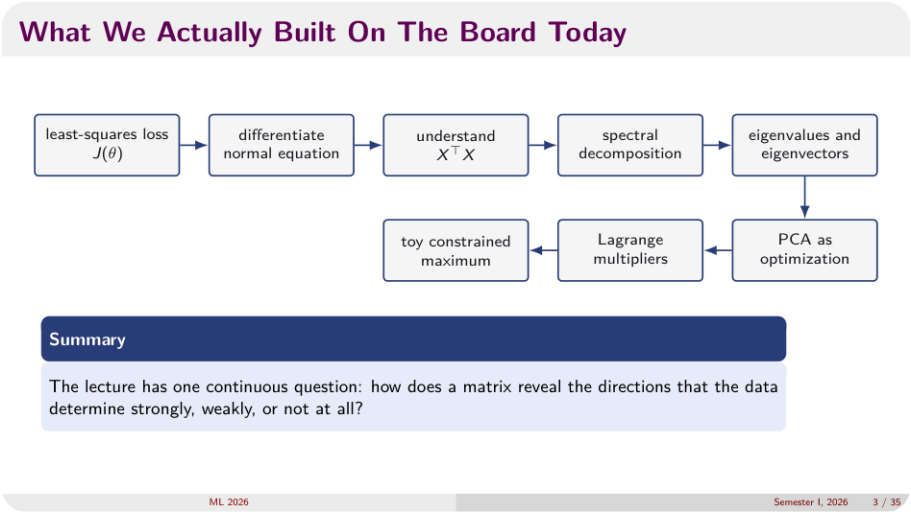
Lecture 5: From the Normal Equation to PCA.

Slide 2 Course Roadmap: You Are Here

Still in **L3–L6 estimation and linear models**. Lecture 4B established least squares and projection. Today:

1. reconstruct the normal equation carefully;
2. use **spectral decomposition** to expose its coefficient directions;
3. reuse the same eigenvalue language to formulate **PCA as constrained optimisation**.

Slide 3 What We Actually Built On The Board Today



Slide 3 — Today's chain from least squares to PCA.

least-squares loss $J(\theta)$ → differentiate → normal equation → understand $X^\top X$ → spectral decomposition → eigenvalues and eigenvectors → PCA as optimisation → Lagrange multipliers → toy constrained maximum.

> **Summary:** one continuous question: *how does a matrix reveal the directions that the data determine strongly, weakly, or not at all?*

Slide 4 Learning Objectives

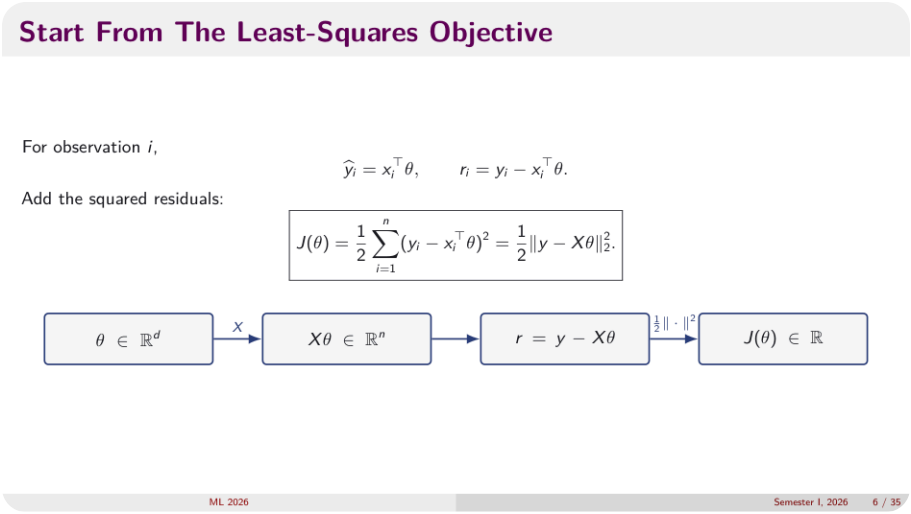
1. Derive $\nabla J(\theta) = X^\top(X\theta - y)$ without guessing the transpose (slides 8–9, A2).
2. Obtain and interpret the normal equation $X^\top X\hat{\theta} = X^\top y$ (slides 9–10).
3. Explain why $X^\top X$ is symmetric and PSD (slides 12–13).
4. Use the spectral theorem to describe regression direction by direction (slides 17–21).
5. Explain where $a = Q^\top\hat{\theta}$ and $b = Q^\top X^\top y$ come from (slides 19–20).
6. Formulate PCA by maximising projected variance (slides 27–29).
7. Derive a Lagrange-multiplier system and reuse it to obtain $Cq = \lambda q$ (slides 30–32).

Slide 5 Notation: Keep Track Of The Space Before The Algebra

Object	Shape	Meaning	Space (A1)
X	$n \times d$	n observations, d predictors	bridge
y	$n \times 1$	observed response	observation
θ	$d \times 1$	regression coefficients	coefficient
$X\theta$	$n \times 1$	fitted response	observation
$r = y - X\theta$	$n \times 1$	residual vector	observation
$X^\top X$	$d \times d$	predictor Gram matrix	acts on coefficient space
q_j	$d \times 1$	one eigenvector direction (which matrix: from context)	coefficient / feature
X_c	$n \times d$	centred feature matrix for PCA	bridge
q	$d \times 1$	one candidate PCA feature direction	feature
$z = X_c q$	$n \times 1$	projected coordinate (score) for every observation	observation
$C = \frac{1}{n-1} X_c^\top X_c$	$d \times d$	sample covariance matrix	acts on feature space

Summary: regression moves a coefficient vector θ. PCA chooses a feature direction q and produces scores $z = X_c q$. **The direction and the scores live in different spaces** ($\mathbb{R}^d$ vs $\mathbb{R}^n$).

Slide 6 Start From The Least-Squares Objective



Slide 6 — θ lives in coefficient space ($\mathbb{R}^d$); Xθ and r live in observation space ($\mathbb{R}^n$).

For observation i: $\hat{y}_i = x_i^\top \theta$, $r_i = y_i - x_i^\top \theta$ (here $x_i^\top$ is **row** i of X). Add squared residuals:

$$J(\theta) = \frac{1}{2}\sum_{i=1}^{n}(y_i - x_i^\top \theta)^2 = \frac{1}{2}\|y - X\theta\|_2^2.$$

Pipeline: $\theta \in \mathbb{R}^d \xrightarrow{X} X\theta \in \mathbb{R}^n \to r = y - X\theta \xrightarrow{\frac{1}{2}\|\cdot\|^2} J(\theta) \in \mathbb{R}$.

(The $\frac{1}{2}$ is only there to cancel the 2 from differentiating; it doesn't move the argmin.)

Slide 7 Expand The Objective Before Differentiating

$$2J(\theta) = (y - X\theta)^\top (y - X\theta) = y^\top y - y^\top X\theta - \theta^\top X^\top y + \theta^\top X^\top X\theta.$$

The two middle terms are the **same scalar** (1×1 equals its own transpose):

$$y^\top X\theta = (y^\top X\theta)^\top = \theta^\top X^\top y.$$

Therefore

$$J(\theta) = \tfrac{1}{2}y^\top y - \theta^\top X^\top y + \tfrac{1}{2}\theta^\top X^\top X\theta.$$

Summary: the transpose appears because the derivative must return a d-dimensional coefficient-space vector. ($X^\top y$ is $d \times 1$; $y^\top X$ would be $1 \times d$.)

Check on L4A data at $\theta = (2.2, 2.2)$: $y^\top y = 4 + 25 + 36 + 81 = 146$; $\theta^\top X^\top y = 2.2(22 + 44) = 145.2$; $\theta^\top X^\top X\theta = 2.2^2(4 + 6 + 6 + 14) = 4.84 \times 30 = 145.2$. $J = 73 - 145.2 + 72.6 = 0.4 = \frac{1}{2}\text{SSE}$ ✓ (SSE was 0.8).

Slide 8 Differentiate One Coefficient At A Time

Start from

$$J(\theta) = \frac{1}{2}\sum_{i=1}^{n}\left(y_i - \sum_{k=1}^{d} x_{ik}\theta_k\right)^2.$$

Let $r_i(\theta) = y_i - \sum_k x_{ik}\theta_k$. Only the $k = j$ term depends on θ_j:

$$\frac{\partial r_i}{\partial \theta_j} = -x_{ij}.$$

Chain rule ($\frac{d}{du}\frac{1}{2}u^2 = u$):

$$\frac{\partial J}{\partial \theta_j} = \frac{1}{2}\sum_{i=1}^{n} 2r_i \frac{\partial r_i}{\partial \theta_j} = -\sum_{i=1}^{n} x_{ij} r_i = \sum_{i=1}^{n} x_{ij}(x_i^\top \theta - y_i).$$

This is **component j of** $X^\top(X\theta - y)$: column j of X dotted with the vector $X\theta - y$. (A2)

Numbers (L4A, at $\theta = 0$): $X\theta - y = (-2, -5, -6, -9)$. $\partial J/\partial\theta_0 = 1 \cdot (-2 - 5 - 6 - 9) = -22$; $\partial J/\partial\theta_1 = 0(-2) + 1(-5) + 2(-6) + 3(-9) = -44$. So $\nabla J(0) = -X^\top y = (-22, -44)$ ✓: both coefficients should increase.

Slide 9 Stack The Derivatives: The Normal Equation Appears

Stack the d scalar derivatives:

$$\nabla J(\theta) = X^\top(X\theta - y) \in \mathbb{R}^d.$$

At a stationary point $\hat{\theta}$:

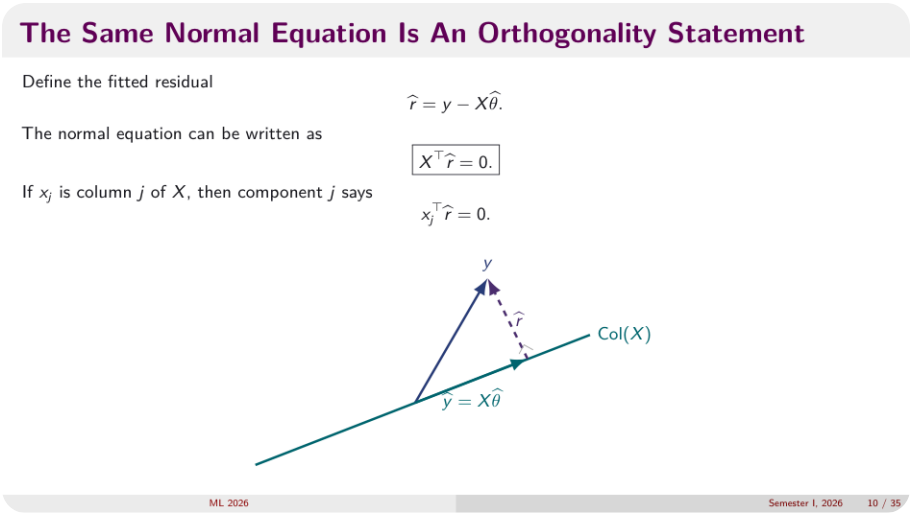
$$X^\top(X\hat{\theta} - y) = 0 \;\Rightarrow\; X^\top X\hat{\theta} - X^\top y = 0 \;\Rightarrow\; \boxed{X^\top X\hat{\theta} = X^\top y}.$$

Dimension check: $X^\top X$ is $d \times d$, $\hat{\theta}$ is $d \times 1$ → left side $d \times 1$. $X^\top$ is $d \times n$, y is $n \times 1$ → right side $d \times 1$ ✓.

Normal equation: one scalar stationarity equation **for each coefficient** (d equations, d unknowns).

Because the Hessian $X^\top X \succeq 0$ (slide 13), every stationary point is a global minimum.

Slide 10 The Same Normal Equation Is An Orthogonality Statement



Slide 10 — The residual is perpendicular to every column of X.

Define the fitted residual $\hat{r} = y - X\hat{\theta}$. The normal equation is

$$X^\top \hat{r} = 0, \qquad \text{i.e. } x_j^\top \hat{r} = 0 \;\text{ for every column } j.$$

The residual is **perpendicular to every predictor column**, hence to all of $\mathrm{Col}(X)$. Picture: y above the plane $\mathrm{Col}(X)$, $\hat{y} = X\hat{\theta}$ its foot, $\hat{r}$ the vertical drop. (L4 slide 28.)

Check (L4A): $\hat{r} = (-0.2, 0.6, -0.6, 0.2)$. Ones column: sum $= 0$ ✓. x column: $0 + 0.6 - 1.2 + 0.6 = 0$ ✓.

Two readings of one equation: calculus says "gradient is zero"; geometry says "residual $\perp$ model space".

Slide 11 When May We Write The Inverse Formula?

From $X^\top X\hat{\theta} = X^\top y$ we may write

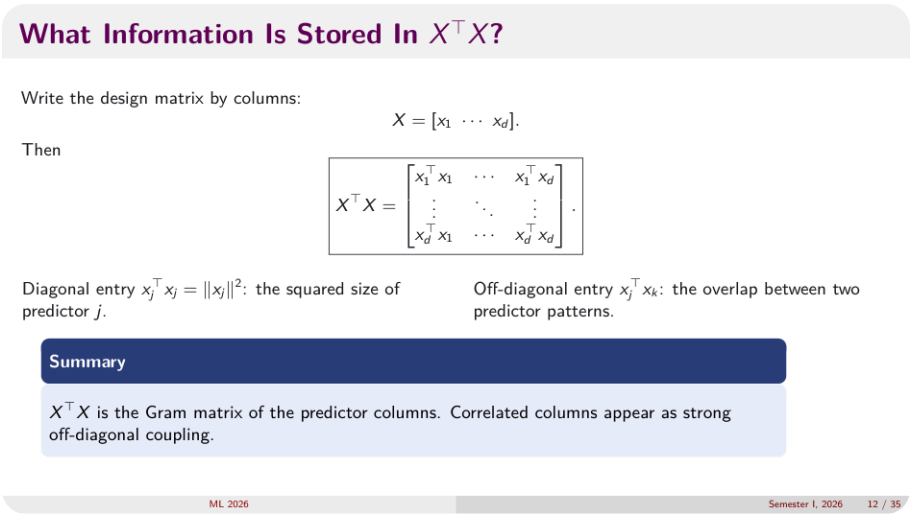
$$\hat{\theta} = (X^\top X)^{-1} X^\top y$$

only when $X^\top X$ is invertible.

Full column rank	Exact linear dependence
$\operatorname{rank}(X) = d$	some predictor is an exact combination of others
no nonzero coefficient move v satisfies $Xv = 0$	$\exists v \neq 0$ with $Xv = 0$
$X^\top X$ PD, all $\lambda_j > 0$	$X^\top X$ singular, some $\lambda_j = 0$
least-squares coefficients **unique**	normal equation still holds (has infinitely many solutions), but the **inverse formula doesn't exist**

Careful. "Solve the normal equation" is always valid. "Invert $X^\top X$" isn't. (In code, use a solver such as `np.linalg.lstsq` / `solve`, not `inv`.)

Slide 12 What Information Is Stored In $X^\top X$?



Slide 12 — Diagonal = column lengths², off-diagonal = overlaps between columns.

Write $X = [x_1 \ \cdots \ x_d]$ by columns. Then

$$X^\top X = \begin{bmatrix} x_1^\top x_1 & \cdots & x_1^\top x_d \\ \vdots & \ddots & \vdots \\ x_d^\top x_1 & \cdots & x_d^\top x_d \end{bmatrix}.$$

- **Diagonal** $x_j^\top x_j = \|x_j\|^2$: the squared size of predictor j.
- **Off-diagonal** $x_j^\top x_k$: the overlap between two predictor patterns.

> **Summary:** $X^\top X$ is the **Gram matrix** of the predictor columns. Correlated columns appear as strong off-diagonal coupling. (A3)

L4A: $\begin{bmatrix} 4 & 6 \\ 6 & 14 \end{bmatrix}$: $\|\mathbf{1}\|^2 = 4$ (= n), $\|x\|^2 = 0 + 1 + 4 + 9 = 14$, overlap $\mathbf{1}^\top x = \sum x_i = 6$.

Slide 13 Why $X^\top X$ Is Positive Semidefinite

Any coefficient direction $v \in \mathbb{R}^d$:

$$v^\top (X^\top X) v = (Xv)^\top (Xv) = \|Xv\|_2^2 \geq 0 \quad \Rightarrow \quad X^\top X \succeq 0.$$

For least squares, $\nabla^2 J(\theta) = X^\top X$, so J **is convex in** θ.

> **Summary:** equality holds exactly when $Xv = 0$: moving the coefficients along v changes **no fitted value**. Strict positivity for every nonzero v gives a **unique** minimiser.

Eigen-reading (preview): if q is a unit eigenvector, $q^\top X^\top X q = \lambda\, q^\top q = \lambda$, and that equals $\|Xq\|^2 \geq 0$. So **every eigenvalue of** $X^\top X$ **is** ≥ 0, and it measures how much moving along q changes the predictions.

Slide 14 Why Eigenvectors Enter Regression At This Point

Rainfall and wetness predictors are strongly coupled:

$$X^\top X = \begin{bmatrix} 10 & 9.8 \\ 9.8 & 10 \end{bmatrix}.$$

The original coefficient axes (θ_{rain}, θ_{wet}) mix the two effects: you can't tell them apart because they almost always move together.

Question: are there special coefficient combinations that $X^\top X$ acts on **without turning them into a different direction**? Those satisfy

$$(X^\top X)q = \lambda q.$$

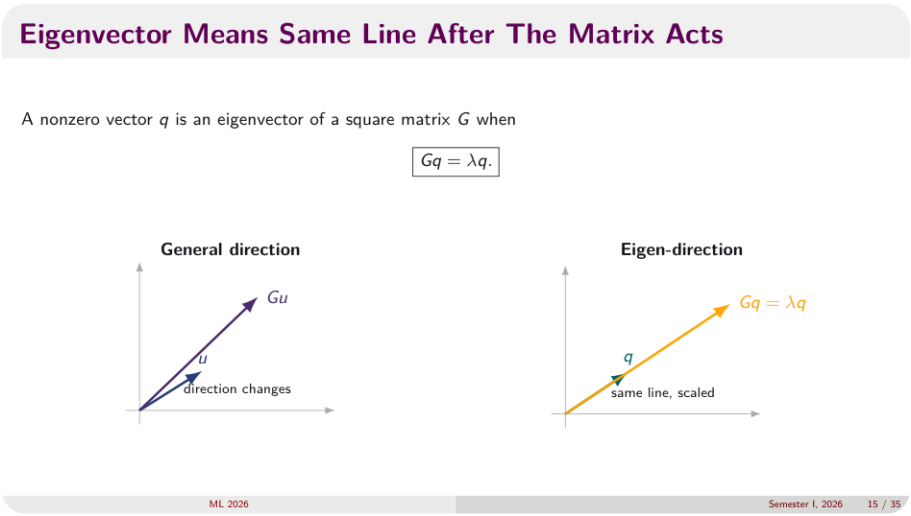
- q: a coefficient combination, e.g. "increase both coefficients together".
- λ: how strongly the data and loss react to that combination.

Answer here (A4 shortcut, $a \pm b$):

Direction	q	λ	Meaning
"both together"	$\frac{1}{\sqrt{2}}(1, 1)$	$10 + 9.8 = 19.8$	strongly determined by data
"one up, other down"	$\frac{1}{\sqrt{2}}(1, -1)$	$10 - 9.8 = 0.2$	barely determined: the data can't separate rain from wetness

Ratio $19.8/0.2 = 99$ (condition number, A7).

Slide 15 Eigenvector Means Same Line After The Matrix Acts



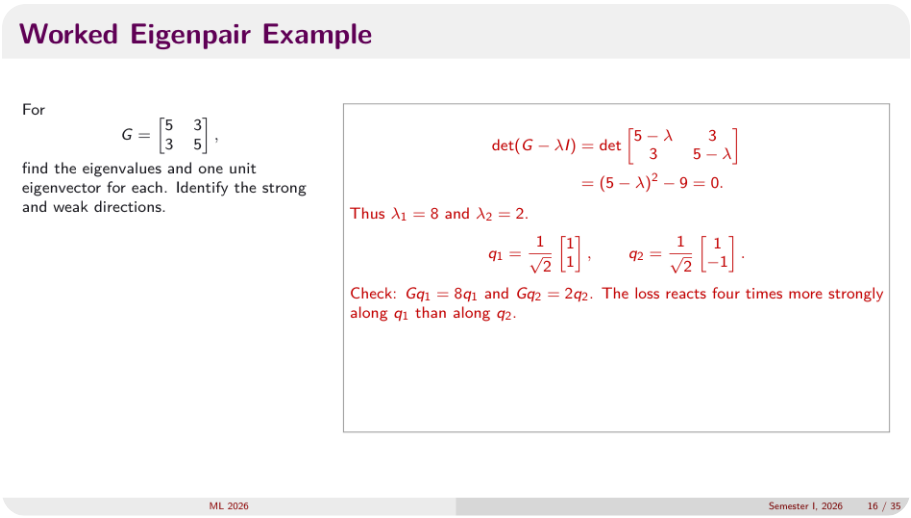
Slide 15 — Left: a general vector gets turned. Right: an eigenvector only gets scaled.

A nonzero q is an eigenvector of a square G when $Gq = \lambda q$.

- **General direction** u: Gu points somewhere else (direction changes).
- **Eigen-direction** q: $Gq = \lambda q$ stays on the **same line**, only scaled.

Example with $G = \begin{bmatrix} 5 & 3 \\ 3 & 5 \end{bmatrix}$: $u = (1,0) \mapsto Gu = (5,3)$, turned. $q = (1,1) \mapsto (8,8)$, same line, ×8.

Slide 16 Worked Eigenpair Example



Slide 16 — Worked det(G − λI) = 0 → λ = 8, 2.

Find the eigenvalues and unit eigenvectors of $G = \begin{bmatrix} 5 & 3 \\ 3 & 5 \end{bmatrix}$; identify the strong and weak directions.

Characteristic equation:

$$\det(G - \lambda I) = \det \begin{bmatrix} 5-\lambda & 3 \\ 3 & 5-\lambda \end{bmatrix} = (5-\lambda)^2 - 9 = 0 \;\Rightarrow\; 5 - \lambda = \pm 3 \;\Rightarrow\; \lambda_1 = 8,\ \lambda_2 = 2.$$

(Check: $8 + 2 = 10 =$ trace ✓; $8 \times 2 = 16 = 25 - 9 =$ det ✓.)

Eigenvectors. $\lambda = 8$: $(G - 8I)q = \begin{bmatrix} -3 & 3 \\ 3 & -3 \end{bmatrix} q = 0 \Rightarrow q_a = q_b$. $\lambda = 2$: $\begin{bmatrix} 3 & 3 \\ 3 & 3 \end{bmatrix} q = 0 \Rightarrow q_a = -q_b$.

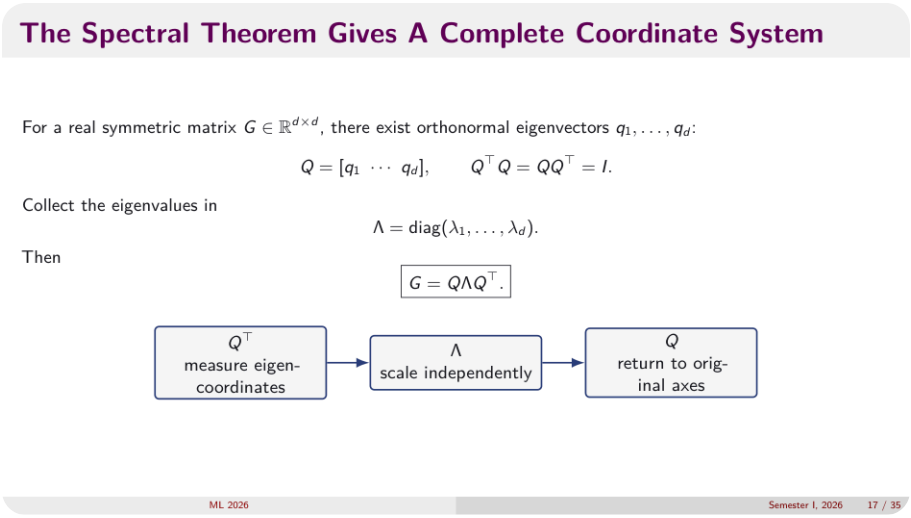
Normalise:

$$q_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad q_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Check: $Gq_1 = \frac{1}{\sqrt{2}}(8,8) = 8q_1$ ✓, $Gq_2 = \frac{1}{\sqrt{2}}(2,-2) = 2q_2$ ✓, $q_1^\top q_2 = \frac{1}{2}(1-1) = 0$ ✓.

Strong direction q_1 ($\lambda = 8$), **weak** direction q_2 ($\lambda = 2$). The loss reacts **four times more strongly** along q_1 than along q_2.

Slide 17 The Spectral Theorem Gives A Complete Coordinate System



Slide 17 — $Q^\top$ rotates in, Λ scales each axis, Q rotates back.

For real symmetric $G \in \mathbb{R}^{d\times d}$ there exist orthonormal eigenvectors $q_1..q_d$:

$$Q = [q_1 \ \cdots \ q_d], \quad Q^\top Q = QQ^\top = I, \quad \Lambda = \mathrm{diag}(\lambda_1..\lambda_d), \quad G = Q\Lambda Q^\top.$$

$Q^\top$: measure eigen-coordinates → Λ: scale independently → Q: return to original axes. (A6)

Check on slide 16: $Q = \frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}$, $\Lambda = \mathrm{diag}(8,2)$.

$$Q\Lambda Q^\top = \tfrac{1}{2}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}\begin{bmatrix}8 & 0\\ 0 & 2\end{bmatrix}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix} = \tfrac{1}{2}\begin{bmatrix}8 & 2\\ 8 & -2\end{bmatrix}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix} = \tfrac{1}{2}\begin{bmatrix}10 & 6\\ 6 & 10\end{bmatrix} = G \checkmark.$$

Or as a sum: $8q_1q_1^\top + 2q_2q_2^\top = 4\begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix} + \begin{bmatrix}1 & -1\\ -1 & 1\end{bmatrix} = \begin{bmatrix}5 & 3\\ 3 & 5\end{bmatrix}$ ✓.

Why symmetric matters: $X^\top X$ and C are symmetric, so this always works for them. (Non-symmetric matrices may have complex eigenvalues or not enough eigenvectors.)

Slide 18 Change Of Basis: The Vector Stays, Its Coordinates Change

Q orthogonal. For a coefficient vector θ, its eigenbasis coordinates are

$$a = Q^\top \theta, \qquad a_j = q_j^\top \theta \ \ (\text{signed coordinate along } q_j).$$

Because $QQ^\top = I$: $\theta = Qa = a_1 q_1 + \cdots + a_d q_d$.

> **Summary:** changing basis does **not** physically rotate the coefficient vector. It changes the axes used to describe it. (A5 has a worked example.)

Interpretation for slide 14: $a_1 = q_1^\top \theta = (\theta_{\text{rain}} + \theta_{\text{wet}})/\sqrt{2}$ is the "combined effect" coordinate; $a_2 = (\theta_{\text{rain}} - \theta_{\text{wet}})/\sqrt{2}$ is the "difference" coordinate.

Slide 19 Rotate The Normal Equation Without Shorthand

Substitute $X^\top X = Q\Lambda Q^\top$ and left-multiply by $Q^\top$. Don't introduce a, b yet.

Line	Reason
$X^\top X\hat{\theta} = X^\top y$	normal equation
$Q\Lambda Q^\top \hat{\theta} = X^\top y$	spectral decomposition
$Q^\top Q\Lambda Q^\top \hat{\theta} = Q^\top X^\top y$	left-multiply **both sides** by $Q^\top$
$\Lambda(Q^\top \hat{\theta}) = Q^\top X^\top y$	$Q^\top Q = I$

Both sides are now written in eigenvector coordinates.

Slide 20 Now Define a And b : Nothing Appears From Nowhere

The previous line is $\Lambda \underbrace{(Q^\top \hat{\theta})}_{a} = \underbrace{Q^\top X^\top y}_{b}$. So

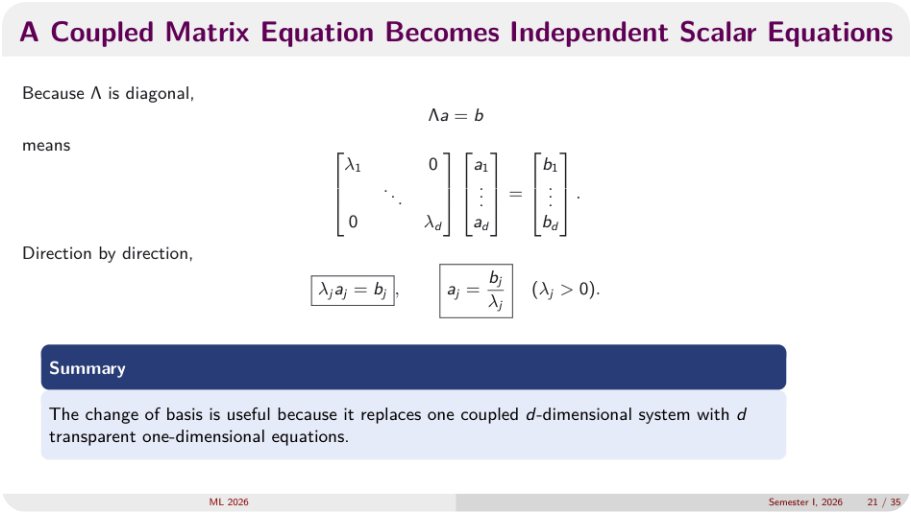
$$a := Q^\top \hat{\theta}, \qquad b := Q^\top X^\top y, \qquad \Lambda a = b.$$

- **Coefficient coordinate:** $a_j = q_j^\top \hat{\theta}$ (the fitted coefficients measured along q_j).

- **Response information in direction q_j:** $b_j = q_j^\top X^\top y = (Xq_j)^\top y$.

b_j measures the **alignment between the observed response y and the prediction pattern Xq_j** generated by moving coefficients along q_j. ($Xq_j \in \mathbb{R}^n$ is "what the predictions would do if you nudged θ along q_j"; dot it with y to see whether the data want that.)

Slide 21 A Coupled Matrix Equation Becomes Independent Scalar Equations



Slide 21 — A coupled 2×2 system becomes separate one-line equations.

Λ is diagonal, so $\Lambda a = b$ is

$$\begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_d \end{bmatrix} \begin{bmatrix} a_1 \\ \vdots \\ a_d \end{bmatrix} = \begin{bmatrix} b_1 \\ \vdots \\ b_d \end{bmatrix} \quad \Longrightarrow \quad \lambda_j a_j = b_j, \qquad a_j = \frac{b_j}{\lambda_j} \ (\lambda_j > 0).$$

Then go back: $\hat{\theta} = Qa = \sum_j \frac{b_j}{\lambda_j} q_j$.

Summary: one coupled d-dimensional system → d transparent 1-D equations.

Worked example (slide 14 matrix). Suppose $X^\top y = (20,\ 19.8)$.

- $b_1 = q_1^\top X^\top y = \frac{39.8}{\sqrt{2}},\ b_2 = q_2^\top X^\top y = \frac{0.2}{\sqrt{2}}$.
- $a_1 = \frac{b_1}{19.8} = \frac{2.0101}{\sqrt{2}},\ a_2 = \frac{b_2}{0.2} = \frac{1}{\sqrt{2}}$.

- $\hat{\theta} = a_1 q_1 + a_2 q_2 = 1.00505\,(1,1) + 0.5\,(1,-1) = (1.505,\ 0.505)$.

Check: $10(1.505) + 9.8(0.505) = 15.05 + 4.95 = 20.0$ ✓; $9.8(1.505) + 10(0.505) = 14.75 + 5.05 = 19.8$ ✓.

Slide 22 Why A Small Eigenvalue Makes Coefficients Sensitive

Perturb $b \to b + \delta b$: $\Lambda(a + \delta a) = b + \delta b$. Subtract $\Lambda a = b$:

$$\Lambda\,\delta a = \delta b \quad \Rightarrow \quad \delta a_j = \frac{\delta b_j}{\lambda_j}.$$

- **Large λ_j:** a fixed data perturbation produces a small coefficient change.
- **Small λ_j:** dividing by a small number **amplifies** it.

Continue the slide-21 example. Change $X^\top y$ from $(20, 19.8)$ to $(20, 20)$ (a 1% nudge in one entry). Now $b_2 = 0$, so $a_2 = 0$, and $\hat{\theta} = \frac{40}{39.6}(1,1) = (1.010,\ 1.010)$.

	$X^\top y$	$\hat{\theta}$
before	$(20,\ 19.8)$	$(1.505,\ 0.505)$
after	$(20,\ 20)$	$(1.010,\ 1.010)$

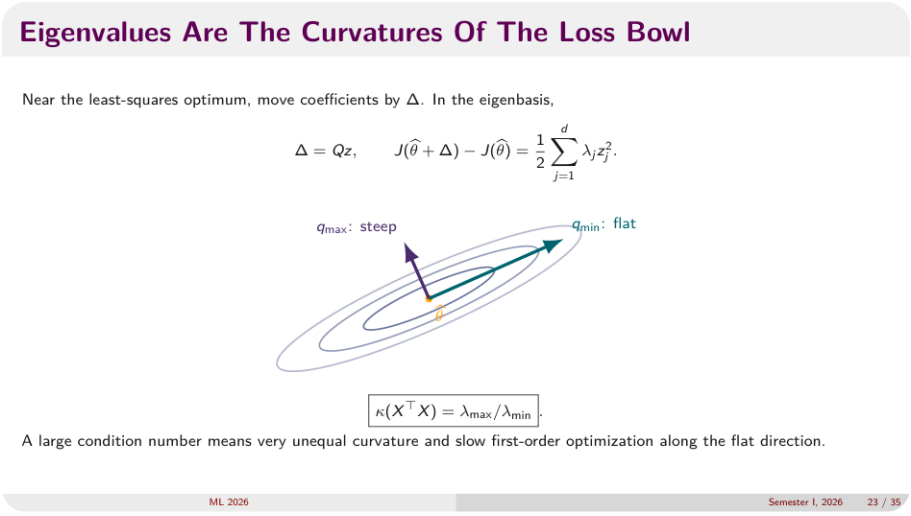
A 0.2 change in the data moved each coefficient by ~0.5: $\delta b_2 = -0.2/\sqrt{2}$ divided by $\lambda_2 = 0.2$ gives $\delta a_2 = -1/\sqrt{2}$. The sum $\theta_1 + \theta_2$ (the strong direction) barely changed: $2.010 \to 2.020$.

> **Summary:** predictions can remain similar, because the same weak direction also satisfies

$$\|Xq_j\|_2^2 = q_j^\top X^\top X q_j = \lambda_j.$$

Moving along a small-λ direction hardly changes $\hat{y}$. That's exactly **why** the data can't pin it down. (This is **multicollinearity**; ridge regression, L7, adds λ_{ridge} to every λ_j to stop the blow-up.)

Slide 23 Eigenvalues Are The Curvatures Of The Loss Bowl



Slide 23 — The bowl is steep along the big-λ direction, flat along the small-λ one.

Near the optimum, move by Δ. Because J is exactly quadratic and $\nabla J(\hat{\theta}) = 0$:

$$J(\hat{\theta}+\Delta) - J(\hat{\theta}) = \underbrace{\nabla J(\hat{\theta})^\top \Delta}_{0} + \tfrac{1}{2}\Delta^\top X^\top X \Delta.$$

In the eigenbasis, $\Delta = Qz$ (A6):

$$J(\hat{\theta}+\Delta) - J(\hat{\theta}) = \frac{1}{2}\sum_{j=1}^{d} \lambda_j z_j^2.$$

Each λ_j is the **curvature** along q_j. Contours are ellipses with axes along the q_j, and the half-length along q_j is $\propto 1/\sqrt{\lambda_j}$: $q_{\min}$ is the long **flat** axis, $q_{\max}$ the short **steep** one.

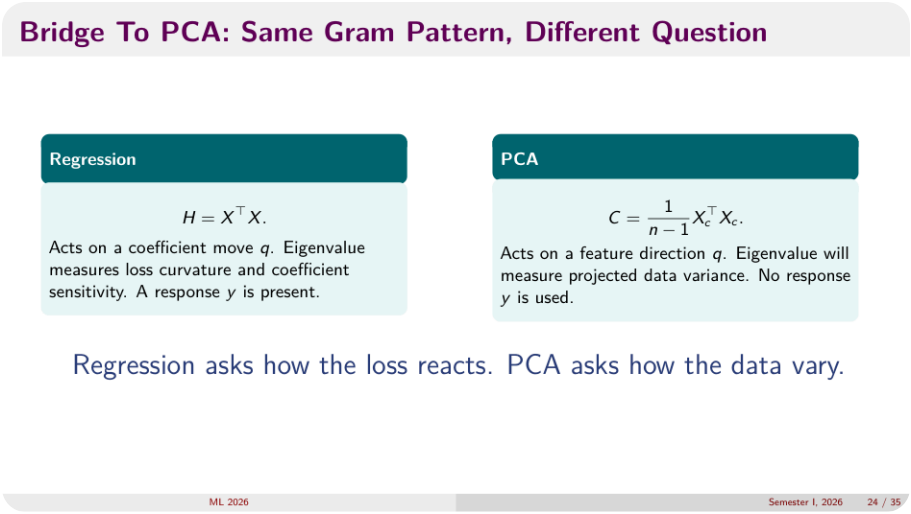
$$\kappa(X^\top X) = \frac{\lambda_{\max}}{\lambda_{\min}}.$$

A large condition number means very unequal curvature and **slow first-order optimisation** (gradient descent) along the flat direction: the step size must be small enough for the steep direction ($< 2/\lambda_{\max}$), so progress along the flat one is tiny. (A7)

Expanding the derivation. $J(\hat{\theta}+\Delta) = \frac{1}{2}\|\hat{r} - X\Delta\|^2 = \frac{1}{2}\|\hat{r}\|^2 - \hat{r}^\top X\Delta + \frac{1}{2}\|X\Delta\|^2$, and $\hat{r}^\top X = 0$ (slide 10).

Slide 24

Bridge To PCA: Same Gram Pattern, Different Question

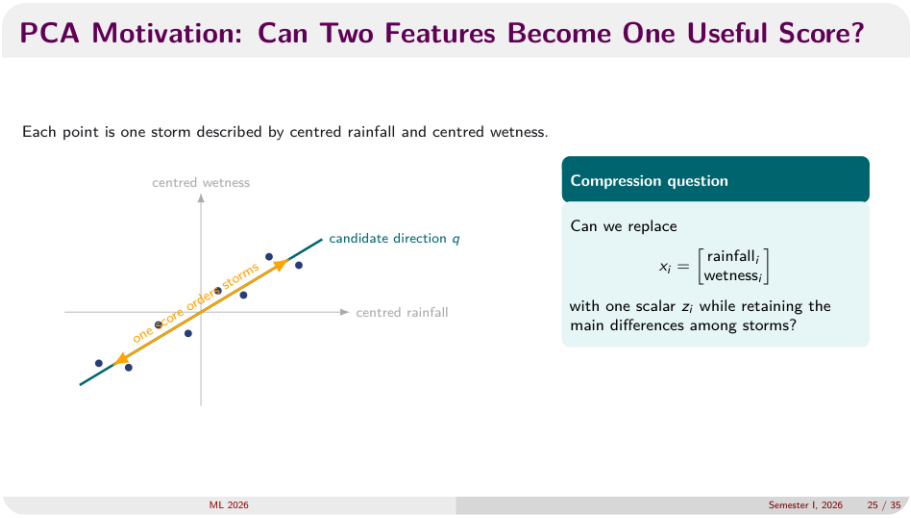


Slide 24 — Same Gram pattern: regression asks how the loss reacts; PCA asks how the data vary.

	Regression	PCA
matrix	$H = X^\top X$	$C = \frac{1}{n-1} X_c^\top X_c$
acts on	a coefficient move q	a feature direction q
eigenvalue measures	loss curvature, coefficient sensitivity	projected data variance
uses a response y?	yes	**no** (unsupervised)

Regression asks how the loss reacts. PCA asks how the data vary.

Slide 25 PCA Motivation: Can Two Features Become One Useful Score?



Slide 25 — The cloud is long along one diagonal; one score along it captures most differences.

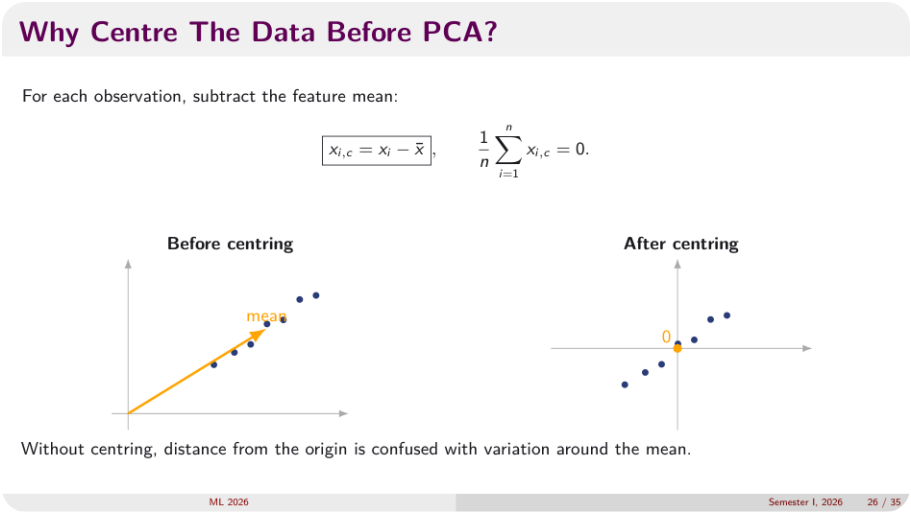
Each point is one storm described by centred rainfall and centred wetness. The cloud is elongated along a diagonal. A candidate direction q through the cloud gives each storm one number (its position along q), and that one score **orders the storms**.

Compression question: can we replace $x_i = \begin{bmatrix} \text{rainfall}_i \\ \text{wetness}_i \end{bmatrix}$ with one scalar z_i while **retaining the main differences among storms**?

(Our toy storms: points $(2, 2), (1, -1), (-1, 1), (-2, -2)$, spread most along the $(1, 1)$ diagonal.)

Slide 26

Why Centre The Data Before PCA?



Slide 26 — Before/after centring: subtract the mean so directions pass through the cloud.

For each observation subtract the feature mean:

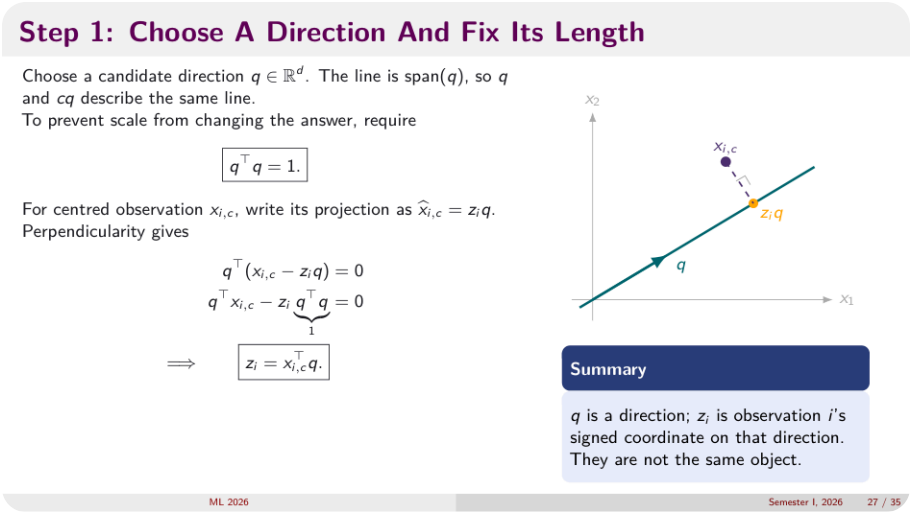
$$x_{i,c} = x_i - \bar{x}, \qquad \frac{1}{n}\sum_{i=1}^{n} x_{i,c} = 0.$$

Before centring the cloud sits around the mean; after, around 0.

Without centring, distance from the origin is confused with variation around the mean. (A8) The direction "towards the cloud's centre" would look like a big-variance direction even if the storms were nearly identical.

Example: raw storms $(52, 32), (51, 29), (49, 31), (48, 28)$. Mean $(50, 30)$. Subtract → our toy table ✓.

Slide 27 Step 1: Choose A Direction And Fix Its Length



Slide 27 — A unit direction q and a point's shadow $z_i = x_i^\top q$ on it.

Choose a candidate $q \in \mathbb{R}^d$. The line is $\mathrm{span}(q)$; q and cq describe the **same line**. To stop scale from changing the answer, require

$$q^\top q = 1.$$

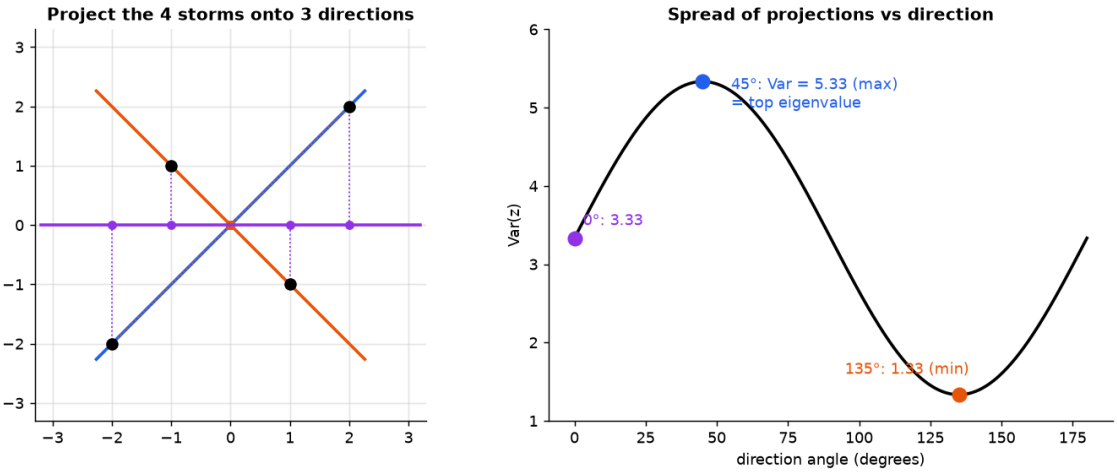
Projection of $x_{i,c}$ onto the line is $\hat{x}_{i,c} = z_i q$. Perpendicularity (A9):

$$q^\top (x_{i,c} - z_i q) = 0 \;\Rightarrow\; q^\top x_{i,c} - z_i \underbrace{q^\top q}_{1} = 0 \;\Rightarrow\; z_i = x_{i,c}^\top q.$$

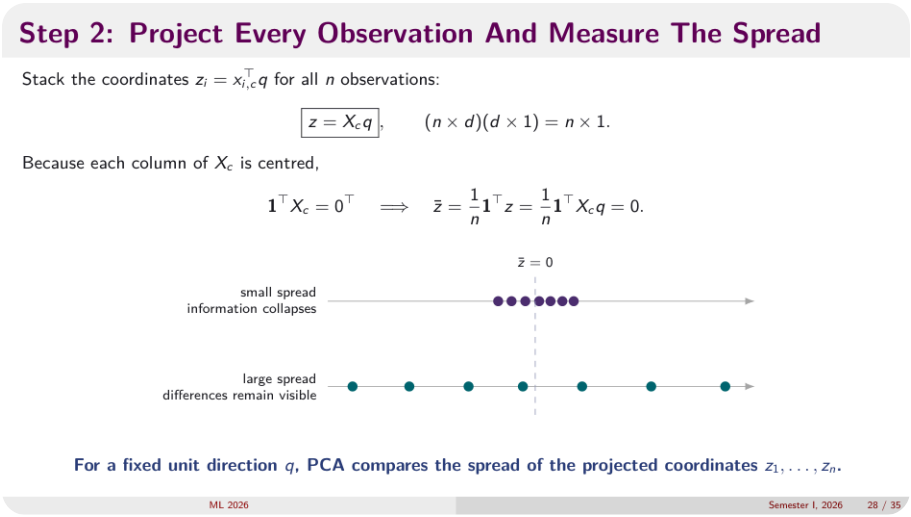
> **Summary:** q is a **direction**; z_i is observation i's **signed coordinate** on that direction. Not the same object.

Toy storms, $q = \frac{1}{\sqrt{2}}(1,1)$: storm 1 $z_1 = \frac{2+2}{\sqrt{2}} = 2\sqrt{2} \approx 2.83$; storm 2 $z_2 = \frac{1-1}{\sqrt{2}} = 0$.

Slide 28 Step 2: Project Every Observation And Measure The Spread



Slide 28's table as a picture: spin the direction, measure the spread; it peaks at 45° (5.33), the top eigenvalue.



Slide 28 — Bunched shadows (small spread) vs spread-out shadows (large spread).

Stack all coordinates:

$$z = X_c q, \qquad (n \times d)(d \times 1) = n \times 1.$$

Because each column of X_c is centred,

$$\mathbf{1}^\top X_c = 0^\top \;\Rightarrow\; \bar{z} = \tfrac{1}{n}\mathbf{1}^\top z = \tfrac{1}{n}\mathbf{1}^\top X_c q = 0.$$

(Centred data ⇒ centred scores, for **any** q.)

Small spread of z	Large spread of z
information collapses: storms land on top of each other	differences remain visible

For a fixed unit q, PCA compares the spread of $z_1..z_n$.

Toy storms, two candidates:

q	$z = X_c q$	$\mathrm{Var}(z) = \frac{z^\top z}{3}$
$\frac{1}{\sqrt{2}}(1,1)$	$\frac{1}{\sqrt{2}}(4,0,0,-4) = (2.83,0,0,-2.83)$	$\frac{8+8}{3} = 5.33$
$\frac{1}{\sqrt{2}}(1,-1)$	$\frac{1}{\sqrt{2}}(0,2,-2,0) = (0,1.41,-1.41,0)$	$\frac{2+2}{3} = 1.33$
$(1,0)$ (rainfall only)	$(2,1,-1,-2)$	$\frac{10}{3} = 3.33$

The diagonal $(1,1)$ keeps far more spread.

Slide 29

Step 3: The Objective And Constraint Now Fit Together

Since $\bar{z} = 0$:

$$\mathrm{Var}(z) = \frac{1}{n-1}\sum_i z_i^2 = \frac{1}{n-1} z^\top z = \frac{1}{n-1}(X_c q)^\top (X_c q) = q^\top \underbrace{\left(\frac{1}{n-1} X_c^\top X_c\right)}_{C} q.$$

Define the **feature covariance matrix** $C = \frac{1}{n-1} X_c^\top X_c$. The first PCA direction is

$$\boxed{q_1 = \arg\max_{q \in \mathbb{R}^d} q^\top C q \quad \text{subject to} \quad q^\top q = 1.}$$

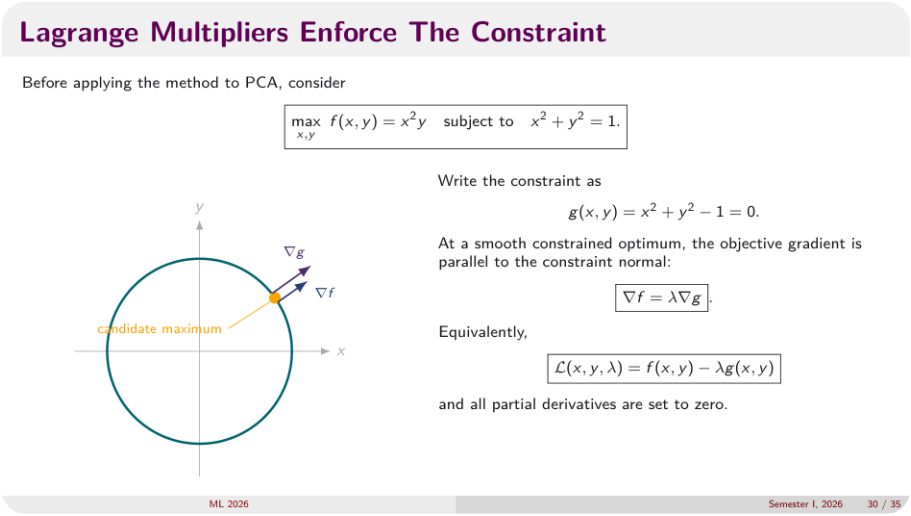
- **Objective** $q^\top C q$: retain the largest projected variance.
- **Constraint** $q^\top q = 1$: prevent scale from manufacturing variance (without it, $q = 10 q_1$ gives 100× the "variance", and the max is ∞).

Toy storms: $X_c^\top X_c = \begin{bmatrix} 4+1+1+4 & 4-1-1+4 \\ 6 & 4+1+1+4 \end{bmatrix} = \begin{bmatrix} 10 & 6 \\ 6 & 10 \end{bmatrix}$, so

$$C = \frac{1}{3}\begin{bmatrix} 10 & 6 \\ 6 & 10 \end{bmatrix} = \begin{bmatrix} 3.33 & 2 \\ 2 & 3.33 \end{bmatrix}.$$

Each feature has variance 3.33; covariance 2 (positively correlated, correlation $2/3.33 = 0.6$). Check slide 28: $q = \frac{1}{\sqrt{2}}(1,1)$ gives $q^\top C q = \frac{1}{2}(3.33 + 2 + 2 + 3.33) = 5.33$ ✓.

Slide 30 Lagrange Multipliers Enforce The Constraint



Slide 30 — On the circle, the best point is where ∇f is parallel to ∇g.

Warm-up before PCA:

$$\max_{x,y} f(x,y) = x^2 y \quad \text{subject to} \quad x^2 + y^2 = 1.$$

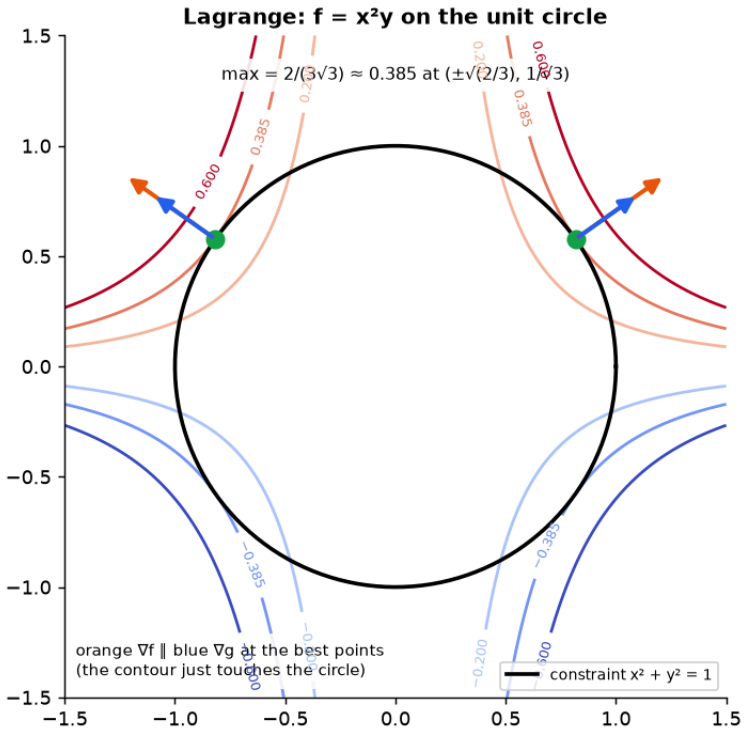
Constraint as $g(x,y) = x^2 + y^2 - 1 = 0$. At a smooth constrained optimum, the objective gradient is **parallel to the constraint normal**:

$$\nabla f = \lambda \nabla g.$$

Equivalently form $\mathcal{L}(x,y,\lambda) = f(x,y) - \lambda g(x,y)$ and set **all** partials to zero. (A10)

Picture: walking around the unit circle, at the best point ∇f points straight out of the circle (along ∇g). If it had any component along the circle, you could walk that way and do better.

Slide 31 Solve $\max x^2y$ On The Unit Circle Step By Step



The best points are where a contour of x²y just touches the circle; there ∇f and ∇g line up.

Solve $\max x^2y$ On The Unit Circle Step By Step

For

$$\mathcal{L} = x^2y - \lambda(x^2 + y^2 - 1),$$

derive the stationary equations, separate the $x = 0$ and $x \neq 0$ cases, and identify the global maximum.

$$\frac{\partial \mathcal{L}}{\partial x} = 2xy - 2\lambda x = 0,$$
$$\frac{\partial \mathcal{L}}{\partial y} = x^2 - 2\lambda y = 0,$$
$$\frac{\partial \mathcal{L}}{\partial \lambda} = -(x^2 + y^2 - 1) = 0.$$

If $x = 0$, then $y = \pm 1$ and $f = 0$.
If $x \neq 0$, the first equation gives $\lambda = y$. The second gives $x^2 = 2y^2$. With $x^2 + y^2 = 1$,

$$3y^2 = 1, \quad y = \pm\frac{1}{\sqrt{3}}, \quad x = \pm\sqrt{\frac{2}{3}}.$$

The global maximum is

$$\boxed{f_{\max} = \frac{2}{3\sqrt{3}}}$$

at $y = 1/\sqrt{3}$ and $x = \pm\sqrt{2/3}$.

ML 2026 Semester I, 2026 31 / 35

Slide 31 — The worked max of x²y on the unit circle.

$$\mathcal{L} = x^2y - \lambda(x^2 + y^2 - 1).$$

Stationary equations:

$$\frac{\partial \mathcal{L}}{\partial x} = 2xy - 2\lambda x = 0, \qquad \frac{\partial \mathcal{L}}{\partial y} = x^2 - 2\lambda y = 0, \qquad \frac{\partial \mathcal{L}}{\partial \lambda} = -(x^2 + y^2 - 1) = 0.$$

The first factorises: $2x(y - \lambda) = 0$, so **either** $x = 0$ **or** $\lambda = y$. (Don't divide by x without splitting cases.)

Case $x = 0$**:** constraint $\Rightarrow y = \pm 1$; second equation $\Rightarrow 0 = 2\lambda y \Rightarrow \lambda = 0$. Then $f = 0$.

Case $x \neq 0$**:** $\lambda = y$. Second equation: $x^2 = 2\lambda y = 2y^2$. Constraint: $2y^2 + y^2 = 1 \Rightarrow y^2 = \frac{1}{3}$:

$$y = \pm\frac{1}{\sqrt{3}}, \qquad x = \pm\sqrt{\tfrac{2}{3}}.$$

Evaluate every candidate:

x	y	$f = x^2 y$
0	± 1	0
$\pm\sqrt{2/3}$	$+1/\sqrt{3}$	$\frac{2}{3} \cdot \frac{1}{\sqrt{3}} = \frac{2}{3\sqrt{3}} \approx 0.385$
$\pm\sqrt{2/3}$	$-1/\sqrt{3}$	$-\frac{2}{3\sqrt{3}} \approx -0.385$

Global maximum $f_{\max} = \dfrac{2}{3\sqrt{3}}$ at $y = 1/\sqrt{3}$, $x = \pm\sqrt{2/3}$. (The negative ones are the global minima.)

Check $\nabla f \parallel \nabla g$ at $(\sqrt{2/3}, 1/\sqrt{3})$: $\nabla f = (2xy, x^2) = (2\sqrt{2}/3,\ 2/3)$; $\nabla g = (2x, 2y) = (2\sqrt{2/3},\ 2/\sqrt{3})$. Ratio in both components: $\lambda = \frac{1}{\sqrt{3}} = y$ ✓.

Slide 32 The PCA Constraint Produces An Eigenvalue Equation

Same method on $\max_q q^\top C q$ s.t. $q^\top q = 1$:

$$\mathcal{L}(q, \lambda) = q^\top C q - \lambda(q^\top q - 1).$$

Because C is symmetric, $\nabla_q(q^\top C q) = 2Cq$ and $\nabla_q(q^\top q) = 2q$ (L4 A9):

$$\nabla_q \mathcal{L} = 2Cq - 2\lambda q = 0 \quad \Longrightarrow \quad \boxed{Cq = \lambda q}.$$

Every stationary direction is an eigenvector of C, and the multiplier is its eigenvalue.

Which one wins? At a unit eigenvector q_j:

$$q_j^\top C q_j = q_j^\top (\lambda_j q_j) = \lambda_j \, q_j^\top q_j = \lambda_j.$$

The **variance captured equals the eigenvalue**. So maximising projected variance means

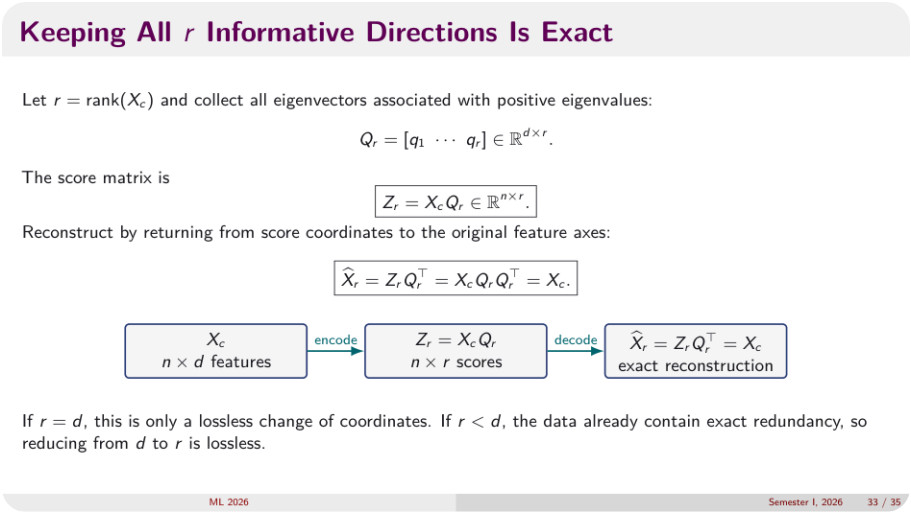
$$q^* = q_1, \qquad \lambda_1 = \max_j \lambda_j.$$

(The Rayleigh bound, A6, confirms no non-eigenvector does better.)

The candidate was named q; after optimisation, the winning direction is the top eigenvector q_1.

Toy storms: $C = \frac{1}{3}\begin{bmatrix} 10 & 6 \\ 6 & 10 \end{bmatrix}$: eigenvalues $\frac{10 \pm 6}{3} = \frac{16}{3}, \frac{4}{3}$ $(5.33, 1.33)$, eigenvectors $\frac{1}{\sqrt{2}}(1,1)$, $\frac{1}{\sqrt{2}}(1,-1)$. Exactly the two variances in the slide-28 table: PC1 = "storm intensity" (rain and wetness together), PC2 = "rain vs wetness contrast".

Slide 33 Keeping All r Informative Directions Is Exact



Slide 33 — Keep all r directions: exact reconstruction, just new coordinates.

Let $r = \operatorname{rank}(X_c)$ and collect the eigenvectors with **positive** eigenvalues: $Q_r = [q_1 \ \cdots \ q_r] \in \mathbb{R}^{d \times r}$.

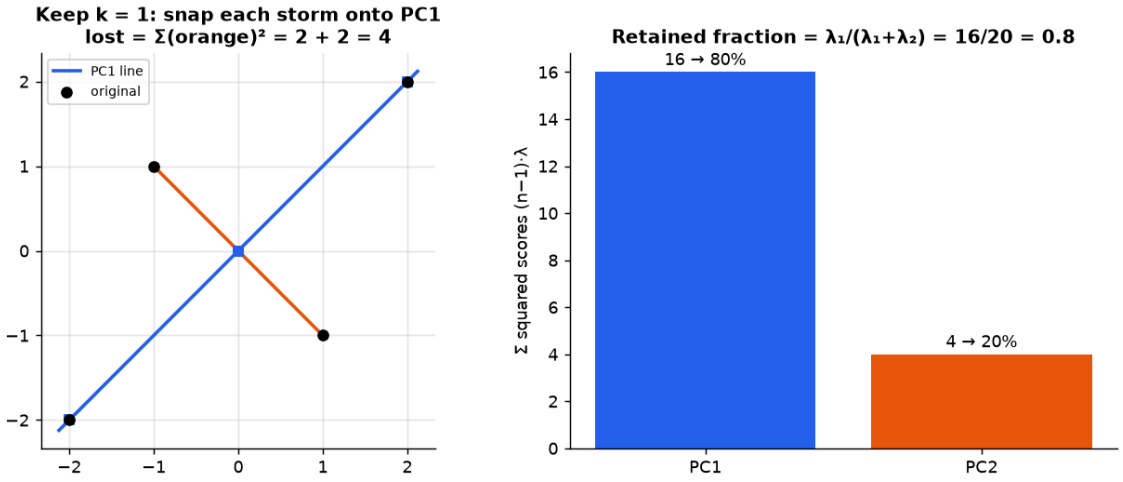
- **Encode** (scores): $Z_r = X_c Q_r \in \mathbb{R}^{n \times r}$.
- **Decode** (back to feature axes): $\hat{X}_r = Z_r Q_r^\top = X_c Q_r Q_r^\top = X_c$.

Why exactly X_c: the dropped directions $q_{r+1}..q_d$ have $\lambda_j = 0$, so $\|X_c q_j\|^2 = (n-1)\lambda_j = 0$, i.e. $X_c q_j = 0$. Then $X_c = X_c \sum_{j=1}^{d} q_j q_j^\top = X_c Q_r Q_r^\top + 0$.

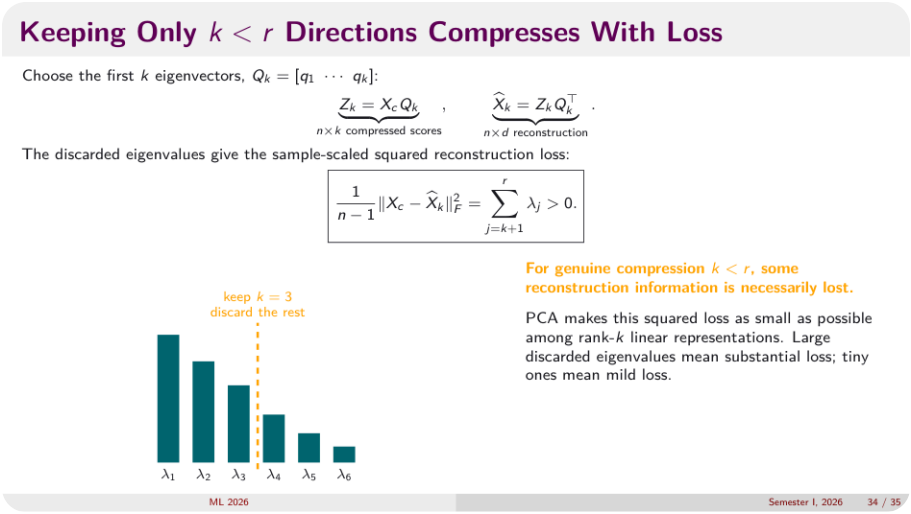
- If $r = d$: just a **lossless change of coordinates** (a rotation).
- If $r < d$: the data already contain **exact redundancy** (some feature is a combination of others), so going from d to r numbers is lossless.

Toy storms: $r = d = 2$ (both eigenvalues positive). $Z_2 = X_c Q$ = rows $(2.83, 0), (0, 1.41), (0, -1.41), (-2.83, 0)$: the same four storms in rotated axes.

Slide 34 Keeping Only $k < r$ Directions Compresses With Loss



Keep 1 of 2 directions: each storm snaps onto PC1; the lost part is the orange gaps (4 of 20, so 80% kept).



Slide 34 — Keep k < r: the bar chart shows how much each dropped λ costs.

Choose the first k eigenvectors $Q_k = [q_1 \cdots q_k]$:

$$\underbrace{Z_k = X_c Q_k}_{n \times k \text{ compressed scores}} \quad , \qquad \underbrace{\hat{X}_k = Z_k Q_k^\top}_{n \times d \text{ reconstruction}} \ .$$

The discarded eigenvalues give the sample-scaled squared reconstruction loss (proof in A11):

$$\frac{1}{n-1} \| X_c - \hat{X}_k \|_F^2 = \sum_{j=k+1}^{r} \lambda_j > 0.$$

Bar-chart picture: $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_6$; keep $k = 3$, discard the rest; the discarded bars' total height is the loss.

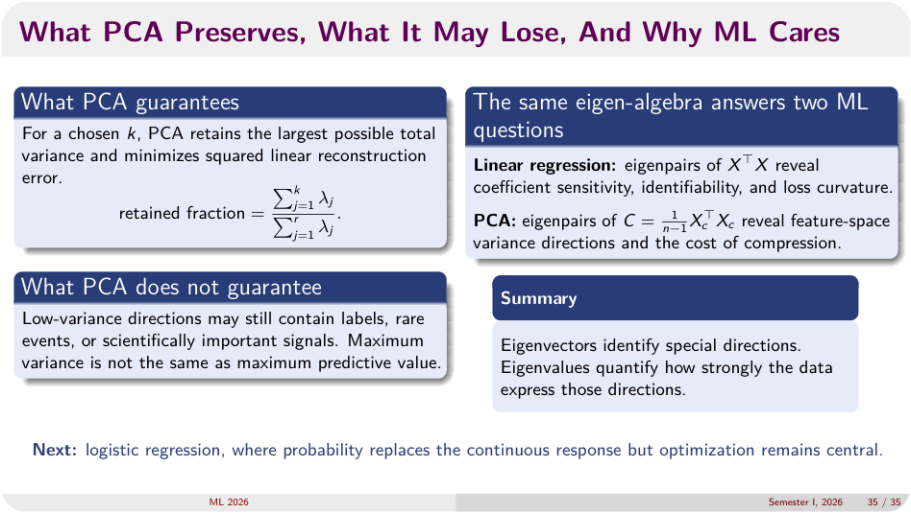
- For genuine compression ($k < r$) some information is **necessarily** lost.
- PCA makes this squared loss **as small as possible** among all rank-k linear representations (you drop the smallest eigenvalues, so the sum you lose is the smallest possible).
- Large discarded eigenvalues = substantial loss; tiny ones = mild loss.

Toy storms, $k = 1$**:** $z = (2.83, 0, 0, -2.83)$, $\hat{X}_1 = z q_1^\top$:

storm	$x_{i,c}$	$\hat{x}_i = z_i q_1$	error
1	$(2, 2)$	$2.83 \cdot \frac{1}{\sqrt{2}}(1, 1) = (2, 2)$	$(0, 0)$
2	$(1, -1)$	$(0, 0)$	$(1, -1)$
3	$(-1, 1)$	$(0, 0)$	$(-1, 1)$
4	$(-2, -2)$	$(-2, -2)$	$(0, 0)$

$\| X_c - \hat{X}_1 \|_F^2 = 1 + 1 + 1 + 1 = 4$; divided by $n - 1 = 3$: $\frac{4}{3} = \lambda_2$ ✓.

Slide 35 What PCA Preserves, What It May Lose, And Why ML Cares



Slide 35 — What PCA guarantees, what it doesn't, and the shared eigen-algebra.

What PCA guarantees. For a chosen k, PCA retains the **largest possible total variance** and **minimises squared linear reconstruction error**.

$$\text{retained fraction} = \frac{\sum_{j=1}^{k} \lambda_j}{\sum_{j=1}^{r} \lambda_j}.$$

(Toy storms, $k = 1$: $\frac{16/3}{20/3} = 80\%$. The denominator is also $\text{trace}(C)$ = total variance $= 3.33 + 3.33$ ✓.)

What PCA does not guarantee. Low-variance directions may still contain **labels, rare events, or scientifically important signals**. Maximum variance ≠ maximum predictive value. (PCA never looks at y. E.g. the small "rain vs wetness contrast" might be exactly what separates flash floods from slow ones.)

Careful. PCA is also **scale-sensitive**: a feature in mm has bigger variance than the same feature in m. Standardise features first if units differ.

The same eigen-algebra answers two ML questions:

	Matrix	Eigenpairs reveal
linear regression	$X^\top X$	coefficient sensitivity, identifiability, loss curvature
PCA	$C = \frac{1}{n-1} X_c^\top X_c$	feature-space variance directions, the cost of compression

> **Summary:** eigenvectors identify **special directions**; eigenvalues quantify **how strongly the data express** them.

Next: logistic regression, where probability replaces the continuous response but optimisation remains central.

Part C Practice

1. For X with columns $\mathbf{1} = (1,1,1)$ and $x = (1,2,3)$, write $X^\top X$. Interpret each entry as a Gram-matrix dot product.
2. Using the componentwise formula $\partial J/\partial\theta_j = \sum_i x_{ij}(x_i^\top\theta - y_i)$, compute ∇J at $\theta = (0,1)$ for the data of Q1 with $y = (2,2,4)$.
3. Find the eigenvalues and unit eigenvectors of $\begin{bmatrix}4 & 1\\ 1 & 4\end{bmatrix}$. Verify $G = Q\Lambda Q^\top$.
4. Find the eigenvalues of $\begin{bmatrix}2 & 1\\ 1 & 3\end{bmatrix}$ with the trace/determinant shortcut. What is the condition number?
5. Show that eigenvectors of a symmetric matrix with different eigenvalues are orthogonal. (Hint: compute $q_1^\top G q_2$ two ways.)
6. With $q_1 = \frac{1}{\sqrt{2}}(1,1)$, $q_2 = \frac{1}{\sqrt{2}}(1,-1)$, find the eigen-coordinates a of $\theta = (5,-1)$ and rebuild θ from them.
7. $X^\top X = \begin{bmatrix}5 & 3\\ 3 & 5\end{bmatrix}$, $X^\top y = (8,4)$. Solve the normal equation **through the eigenbasis** ($b = Q^\top X^\top y$, $a_j = b_j/\lambda_j$, $\hat\theta = Qa$). Check by direct solve.
8. Show $\|Xq_j\|^2 = \lambda_j$ for a unit eigenvector of $X^\top X$. What does $\lambda_j = 0$ mean for predictions?
9. Why can't we write $\hat\theta = (X^\top X)^{-1}X^\top y$ if two predictor columns are identical? Does the normal equation still have a solution?
10. Maximise $f(x,y) = xy$ on $x^2 + y^2 = 1$ with Lagrange multipliers. Which λ goes with the maximum?
11. Points $(3,1), (1,3), (-1,-3), (-3,-1)$ (already centred). Compute C, its eigenpairs, PC1 scores, and the retained-variance fraction for $k = 1$.
12. Why must PCA constrain $q^\top q = 1$? What happens to $\max q^\top C q$ without it?
13. For Q11, compute the $k = 1$ reconstruction error $\frac{1}{n-1}\|X_c - \hat X_1\|_F^2$ directly and compare with λ_2.
14. A colleague drops all but the top 2 PCs before training a classifier and accuracy collapses. Give one reason that could happen.

1. $X^\top X = \begin{bmatrix}3 & 6\\ 6 & 14\end{bmatrix}$: $\mathbf{1}^\top\mathbf{1} = 3 = n$, $\mathbf{1}^\top x = 6 = \sum x_i$, $x^\top x = 14 = \sum x_i^2$.
2. $X\theta = (1,2,3)$, $X\theta - y = (-1,0,-1)$. $\partial/\partial\theta_0 = -1 + 0 - 1 = -2$; $\partial/\partial\theta_1 = 1(-1) + 2(0) + 3(-1) = -4$. $\nabla J = (-2,-4)$.
3. $\lambda = 4 \pm 1 = 5, 3$; $q_1 = \frac{1}{\sqrt{2}}(1,1)$, $q_2 = \frac{1}{\sqrt{2}}(1,-1)$. $5q_1q_1^\top + 3q_2q_2^\top = \frac{5}{2}\begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix} + \frac{3}{2}\begin{bmatrix}1 & -1\\ -1 & 1\end{bmatrix} = \begin{bmatrix}4 & 1\\ 1 & 4\end{bmatrix}$ ✓.

4. $T = 5$, $D = 5$: $\lambda = 2.5 \pm \sqrt{6.25 - 5} = 2.5 \pm 1.118 = 3.618,\ 1.382$. $\kappa = 3.618/1.382 \approx 2.62$.
5. $q_1^\top G q_2 = q_1^\top (\lambda_2 q_2) = \lambda_2 q_1^\top q_2$. Also $q_1^\top G q_2 = (G^\top q_1)^\top q_2 = (G q_1)^\top q_2 = \lambda_1 q_1^\top q_2$ (uses $G^\top = G$). Subtract: $(\lambda_1 - \lambda_2) q_1^\top q_2 = 0$, and $\lambda_1 \neq \lambda_2 \Rightarrow q_1^\top q_2 = 0$.
6. $a_1 = \frac{4}{\sqrt{2}} = 2\sqrt{2}$, $a_2 = \frac{6}{\sqrt{2}} = 3\sqrt{2}$. $2\sqrt{2} q_1 + 3\sqrt{2} q_2 = (2, 2) + (3, -3) = (5, -1)$ ✓.
7. $b_1 = \frac{12}{\sqrt{2}}$, $b_2 = \frac{4}{\sqrt{2}}$. $a_1 = \frac{12}{8\sqrt{2}} = \frac{1.5}{\sqrt{2}}$, $a_2 = \frac{4}{2\sqrt{2}} = \frac{2}{\sqrt{2}}$. $\hat{\theta} = 0.75(1, 1) + 1(1, -1) = (1.75, -0.25)$. Direct: $5(1.75) + 3(-0.25) = 8$ ✓, $3(1.75) + 5(-0.25) = 4$ ✓.
8. $\|X q_j\|^2 = q_j^\top X^\top X q_j = q_j^\top \lambda_j q_j = \lambda_j$. $\lambda_j = 0 \Rightarrow X q_j = 0$: moving coefficients along q_j changes **no** prediction, so the data say nothing about that direction.
9. Identical columns $\Rightarrow Xv = 0$ for $v = (1, -1, 0, \dots) \Rightarrow X^\top X$ singular (eigenvalue 0) $\Rightarrow$ no inverse. The normal equation still has solutions (always: $X^\top y$ lies in $\mathrm{Col}(X^\top) = \mathrm{Col}(X^\top X)$), infinitely many.
10. $\nabla f = (y, x) = \lambda(2x, 2y) \Rightarrow y = 2\lambda x$, $x = 2\lambda y \Rightarrow x = 4\lambda^2 x$. $x = 0$ forces $y = 0$, violating the constraint, so $\lambda = \pm\frac{1}{2}$. $\lambda = \frac{1}{2}$: $x = y = \pm\frac{1}{\sqrt{2}}$, $f = \frac{1}{2}$ (max). $\lambda = -\frac{1}{2}$: $x = -y$, $f = -\frac{1}{2}$ (min). The multiplier equals the optimal value here (like PCA, where λ = variance).
11. $X_c^\top X_c = \begin{bmatrix} 9+1+1+9 & 3+3+3+3 \\ 12 & 1+9+9+1 \end{bmatrix} = \begin{bmatrix} 20 & 12 \\ 12 & 20 \end{bmatrix}$, $C = \frac{1}{3}\begin{bmatrix} 20 & 12 \\ 12 & 20 \end{bmatrix}$. $\lambda = \frac{32}{3}, \frac{8}{3}$; $q_1 = \frac{1}{\sqrt{2}}(1, 1)$, $q_2 = \frac{1}{\sqrt{2}}(1, -1)$. Scores $z = \frac{1}{\sqrt{2}}(4, 4, -4, -4) = (2.83, 2.83, -2.83, -2.83)$. Retained $= \frac{32}{40} = 80\%$.
12. $q^\top C q$ scales as c^2 when $q \to cq$, so without a constraint the max is $+\infty$ (just make q longer). The constraint compares directions, not lengths.
13. $\hat{x}_i = z_i q_1$: $(2, 2), (2, 2), (-2, -2), (-2, -2)$. Errors $(1, -1), (-1, 1), (1, -1), (-1, 1)$, each squared norm 2, total 8. $8/3 = \lambda_2$ ✓.
14. PCA ignores the labels: the class-separating signal may sit in a low-variance direction that got dropped (slide 35). Also unscaled features can let one large-unit feature dominate the top PCs.